Module 07: Perimeter Defence Evasion Techniques

```
# Exercise 03: HTTP Tunneling to Bypass Firewalls Using HTTPort
```

Scenario

Perimeter firewalls and egress filtering policies frequently restrict outbound high-risk protocol ports (such as FTP port 21), while allowing unencrypted HTTP (TCP port 80) traffic to ensure essential web connectivity. Attackers leverage **HTTP Tunneling** techniques to encapsulate restricted protocol streams inside allowed HTTP requests/responses, effectively bypassing strict egress firewall controls and network intrusion prevention systems.

Using **HTTPort** on a restricted client and **HTTHost** on an intermediate gateway server, egress traffic targeting blocked destination services (e.g., FTP server at **172.19.19.17**) can be proxied and encapsulated across standard HTTP sessions listening on port 80 (**172.19.19.9:80**). Configuring personal authentication passwords in HTTHost's **Options** tab (**options**) ensures secure, authenticated tunnel proxying.

In this exercise, we stop IIS web services and run HTTHost on the FTP Server (**172.19.19.9**), configure an outbound Port 21 blocking firewall rule on Windows Server 2022, configure HTTPort proxy and port mapping settings to target Sales Department (**172.19.19.17**), execute localhost FTP connections (**ftp 127.0.0.1**), and verify directory creation through the HTTP tunnel.

Objective

The objective of this lab is to help students learn how to:
- Configure **HTTHost** listener services on remote intermediate servers (**172.19.19.9:80**)
- Identify the personal password configuration interface in HTTHost (**Options** tab $
ightarrow$ **options**)
- Configure Windows Defender Firewall Advanced Outbound Rules to block destination FTP port 21
- Verify egress firewall block enforcement against direct target IPs (**172.19.19.17:21**)
- Install and configure **HTTPort** proxy settings, bypass host parameters, and port mappings
- Tunnel FTP traffic via localhost (**ftp 127.0.0.1**) through HTTPort/HTTHost to authenticate and interact with remote file systems

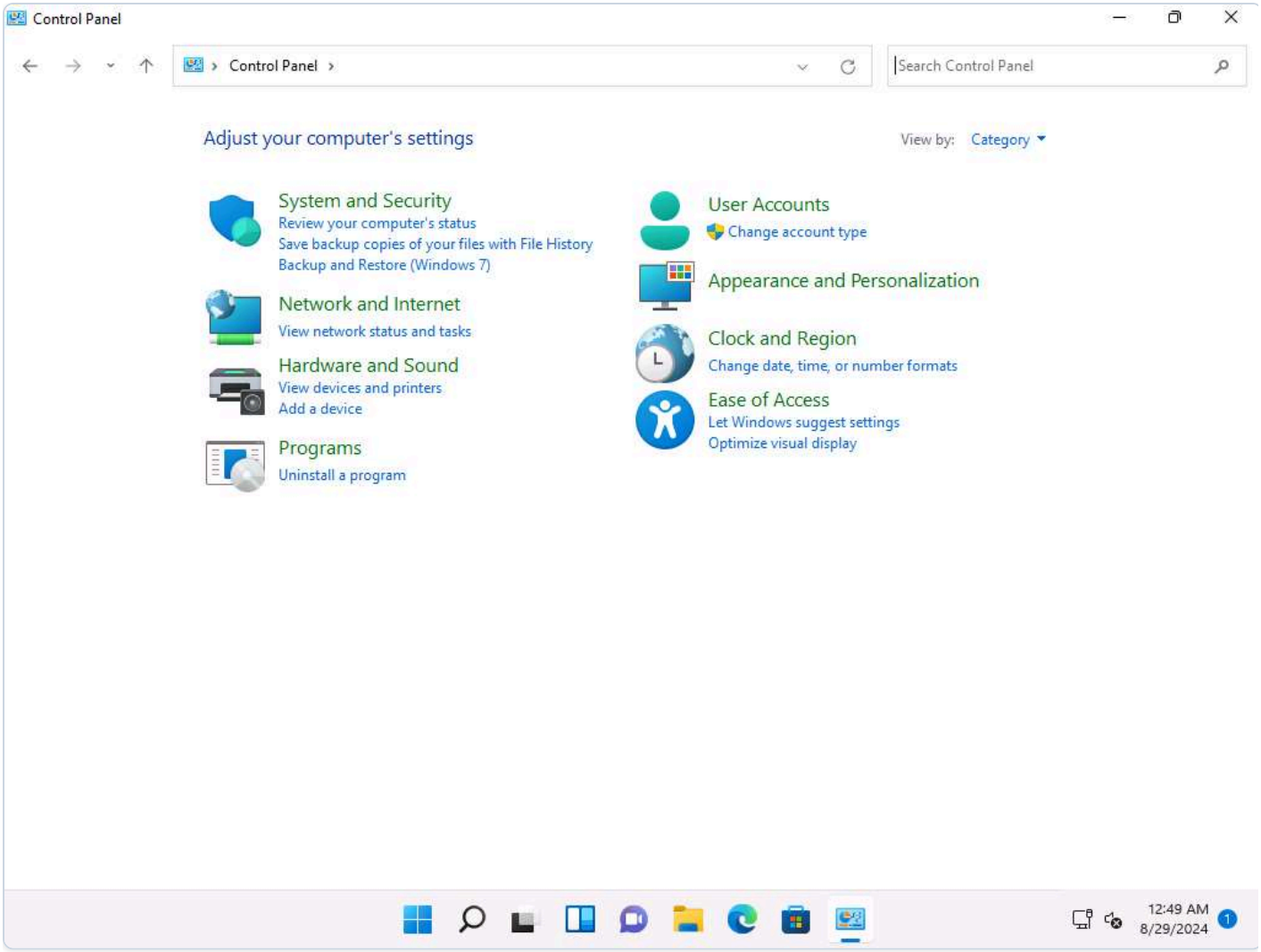
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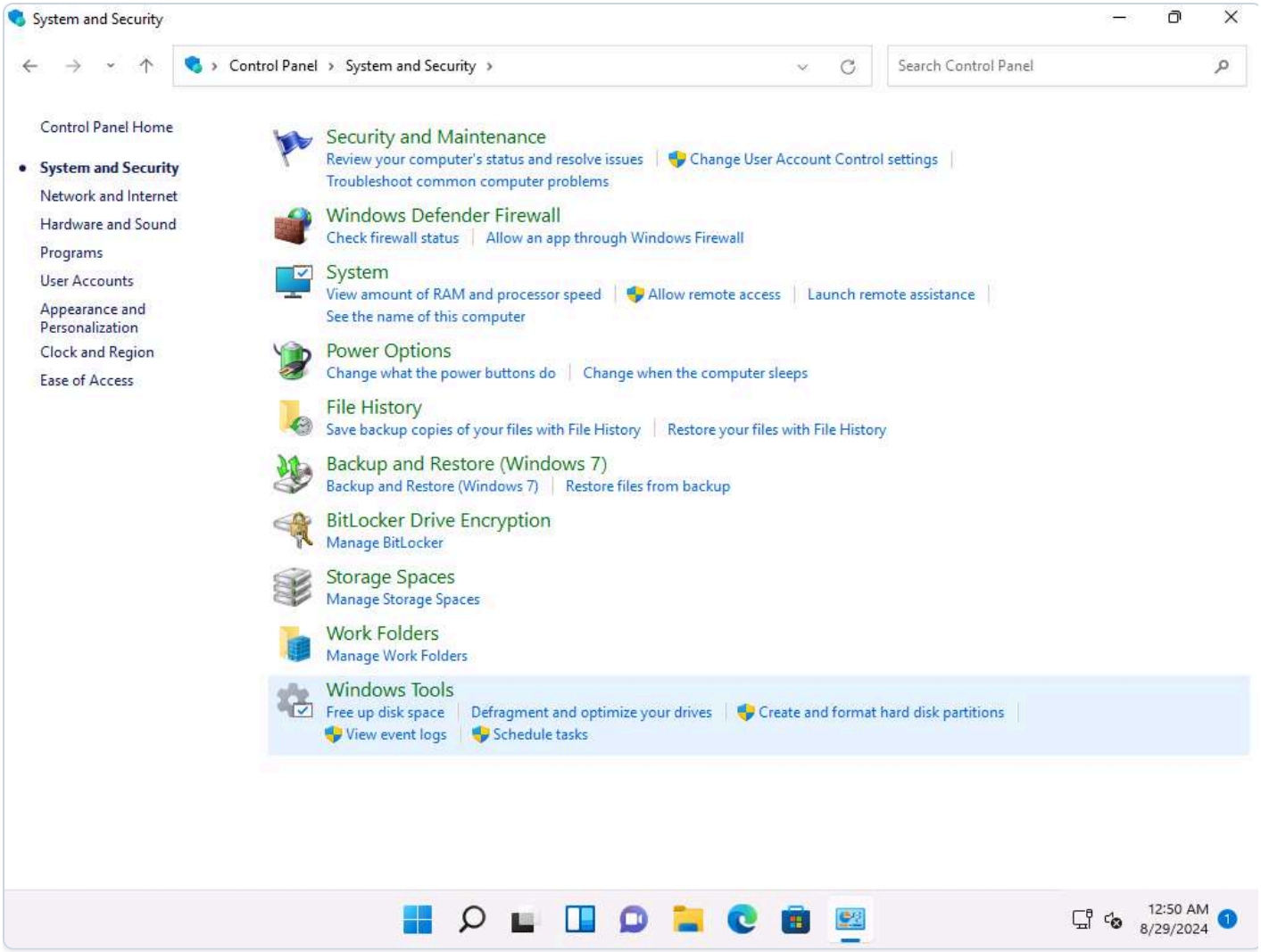
Task 1: Initialize HTTHost Server Listener

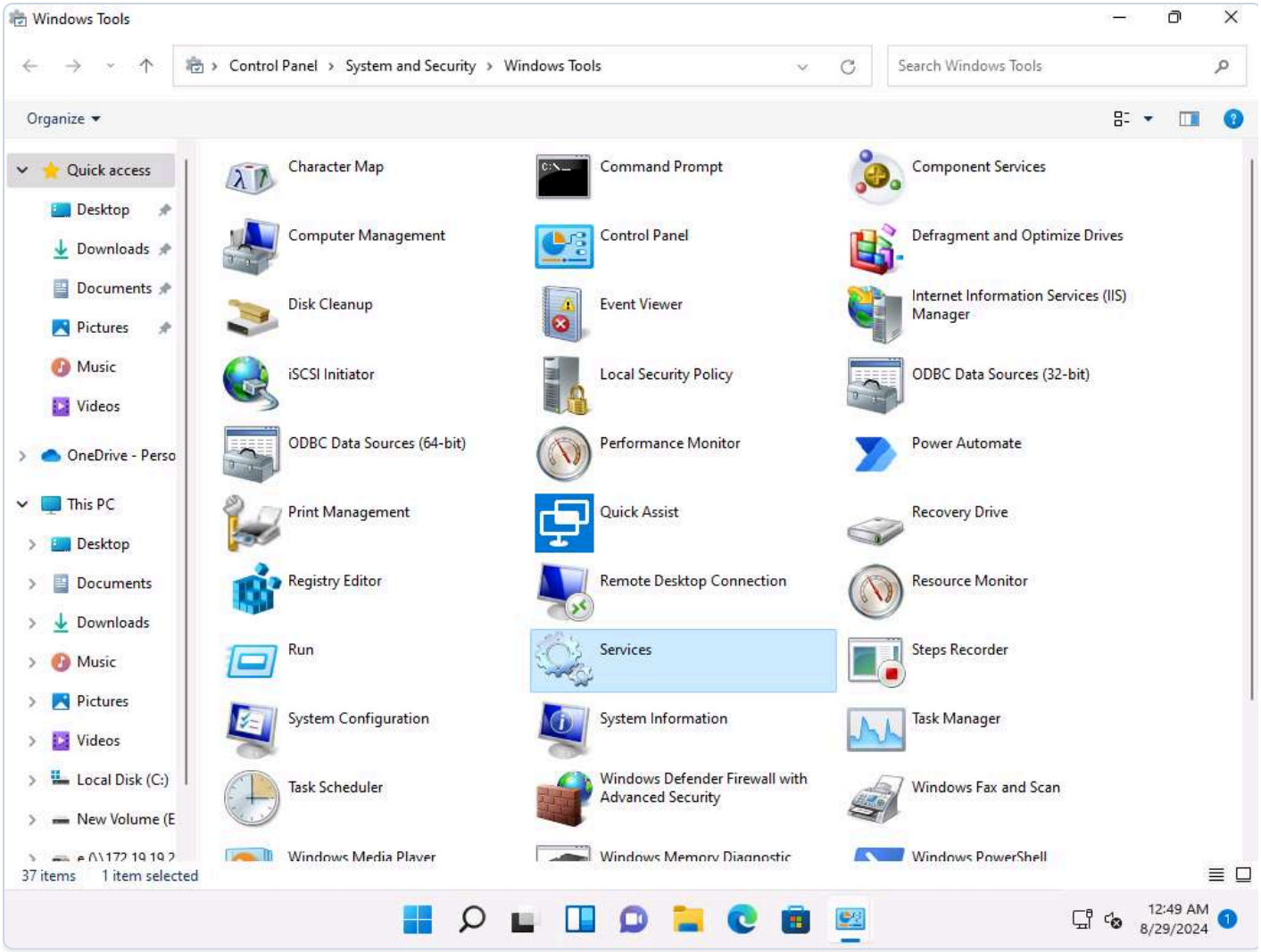
In this task, we stop conflicting IIS web services on the FTP Server machine (**172.19.19.9**), configure HTTHost listener options, set the personal password (**magic**) in the **Options** tab (**options**), and verify HTTP port 80 binding.

Lab Tasks

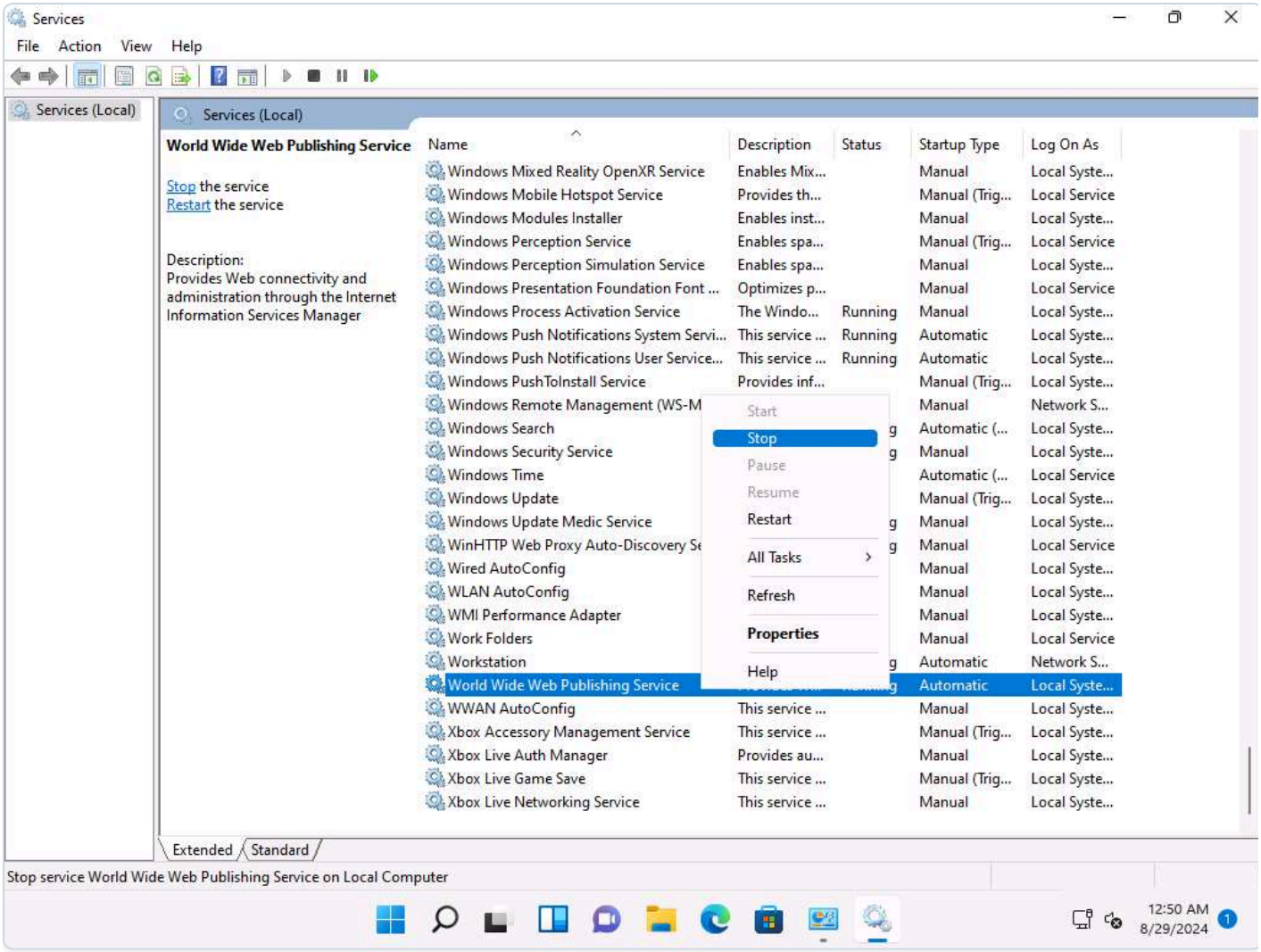
1. Switch to **FTP Server** and press **Ctrl+Alt+Delete**. Log in with password **test@123**.
2. Open **Control Panel** $
   ightarrow$ **System and Security** $
   ightarrow$ **Windows Tools** $
   ightarrow$ **Services**.


Control Panel
Control Panel
Search Control Panel
Adjust your computer's settings
View by: Category
System and Security
Review your computer's status
Save backup copies of your files with File History
Backup and Restore (Windows 7)
Network and Internet
View network status and tasks
Hardware and Sound
View devices and printers
Add a device
Programs
Uninstall a program
User Accounts
Change account type
Appearance and Personalization
Clock and Region
Change date, time, or number formats
Ease of Access
Let Windows suggest settings
Optimize visual display
12:49 AM
8/29/2024

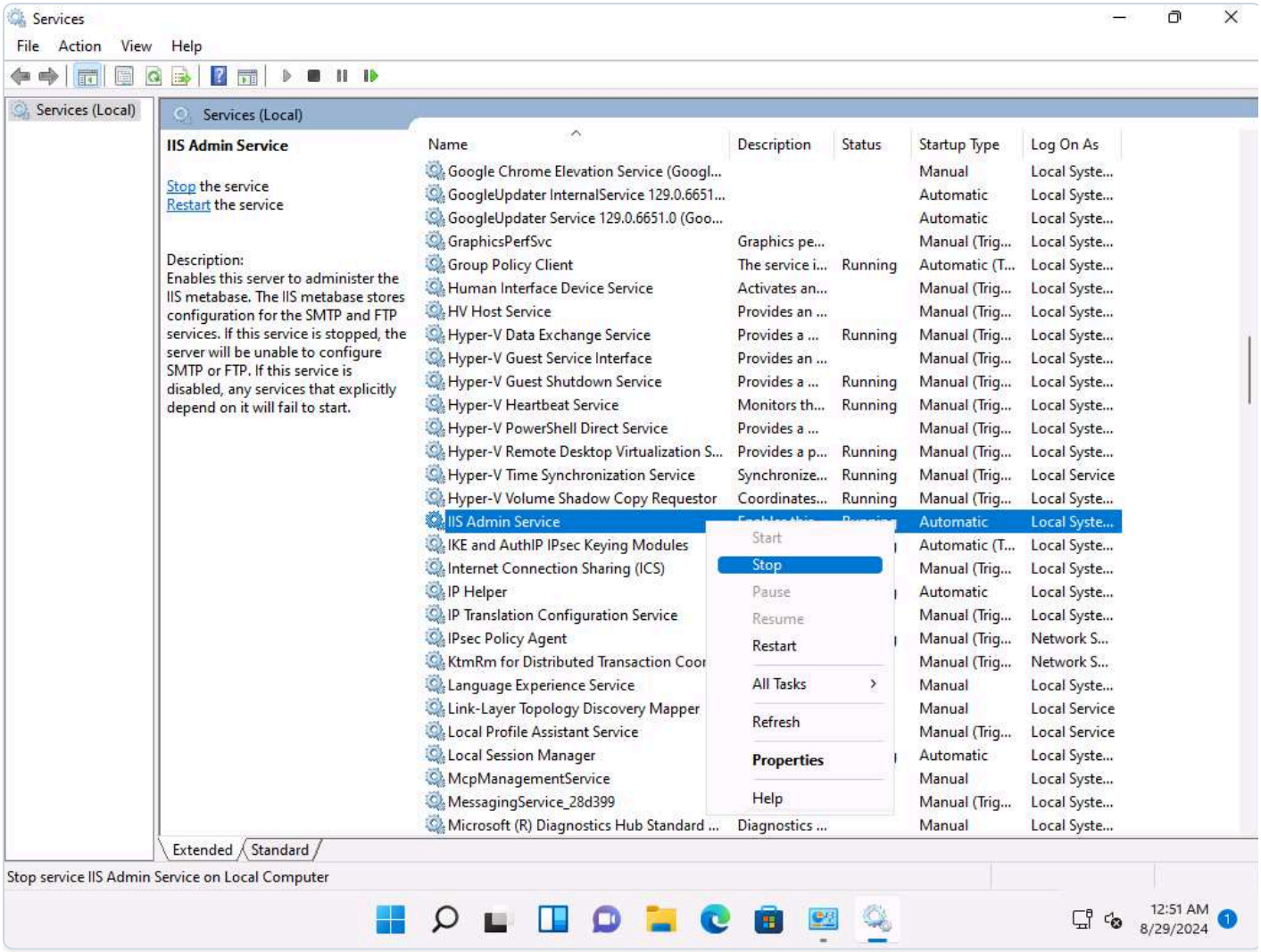

System and Security
Control Panel › System and Security ›
Search Control Panel
Control Panel Home
System and Security
Network and Internet
Hardware and Sound
Programs
User Accounts
Appearance and Personalization
Clock and Region
Ease of Access
Security and Maintenance
Review your computer's status and resolve issues
Change User Account Control settings
Troubleshoot common computer problems
Windows Defender Firewall
Check firewall status
Allow an app through Windows Firewall
System
View amount of RAM and processor speed
Allow remote access
Launch remote assistance
See the name of this computer
Power Options
Change what the power buttons do
Change when the computer sleeps
File History
Save backup copies of your files with File History
Restore your files with File History
Backup and Restore (Windows 7)
Backup and Restore (Windows 7)
Restore files from backup
BitLocker Drive Encryption
Manage BitLocker
Storage Spaces
Manage Storage Spaces
Work Folders
Manage Work Folders
Windows Tools
Free up disk space
Defragment and optimize your drives
Create and format hard disk partitions
View event logs
Schedule tasks
12:50 AM
8/29/2024



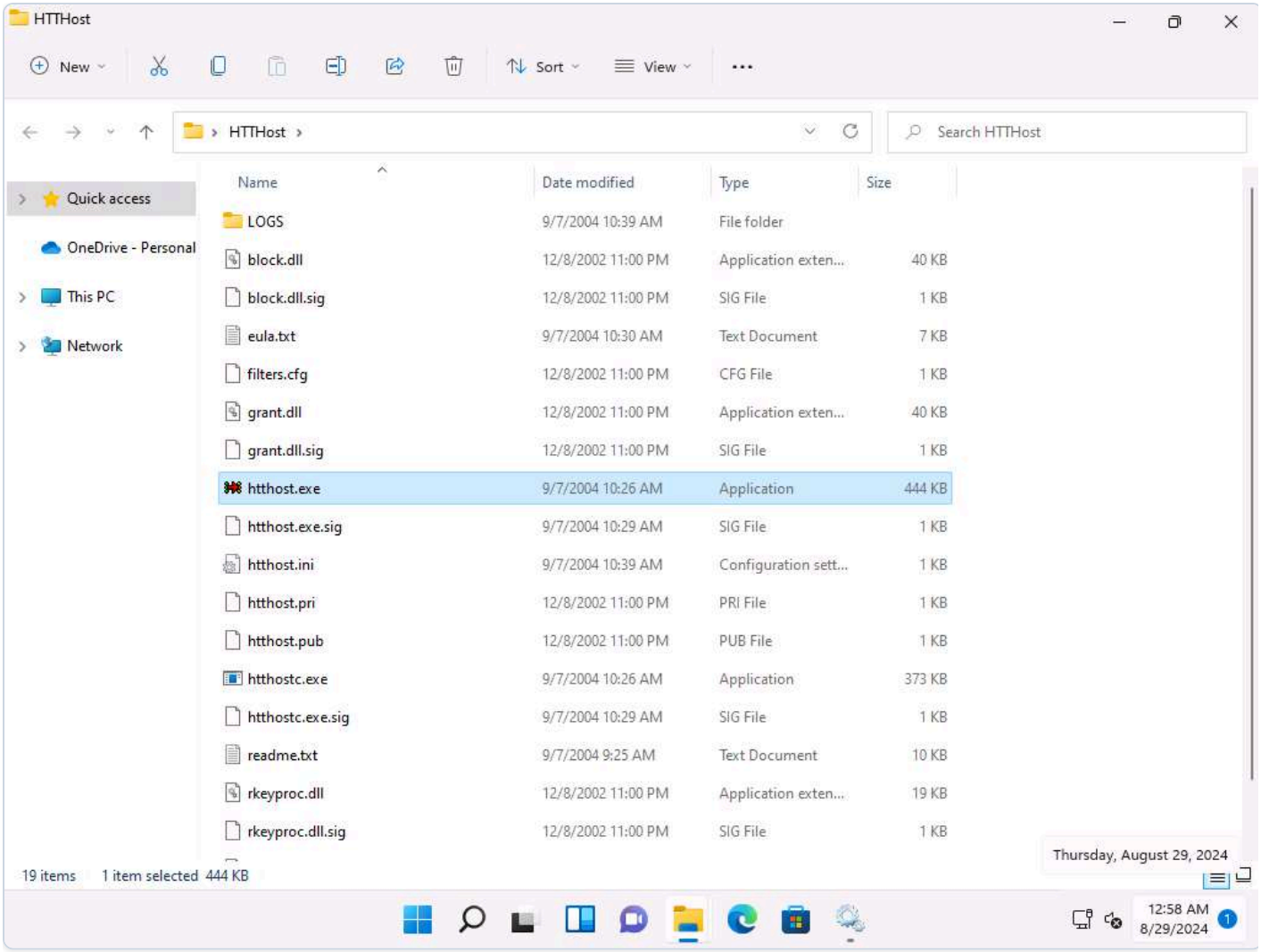
1. Locate **World Wide Web Publishing Service**, right-click, and select **Stop**.

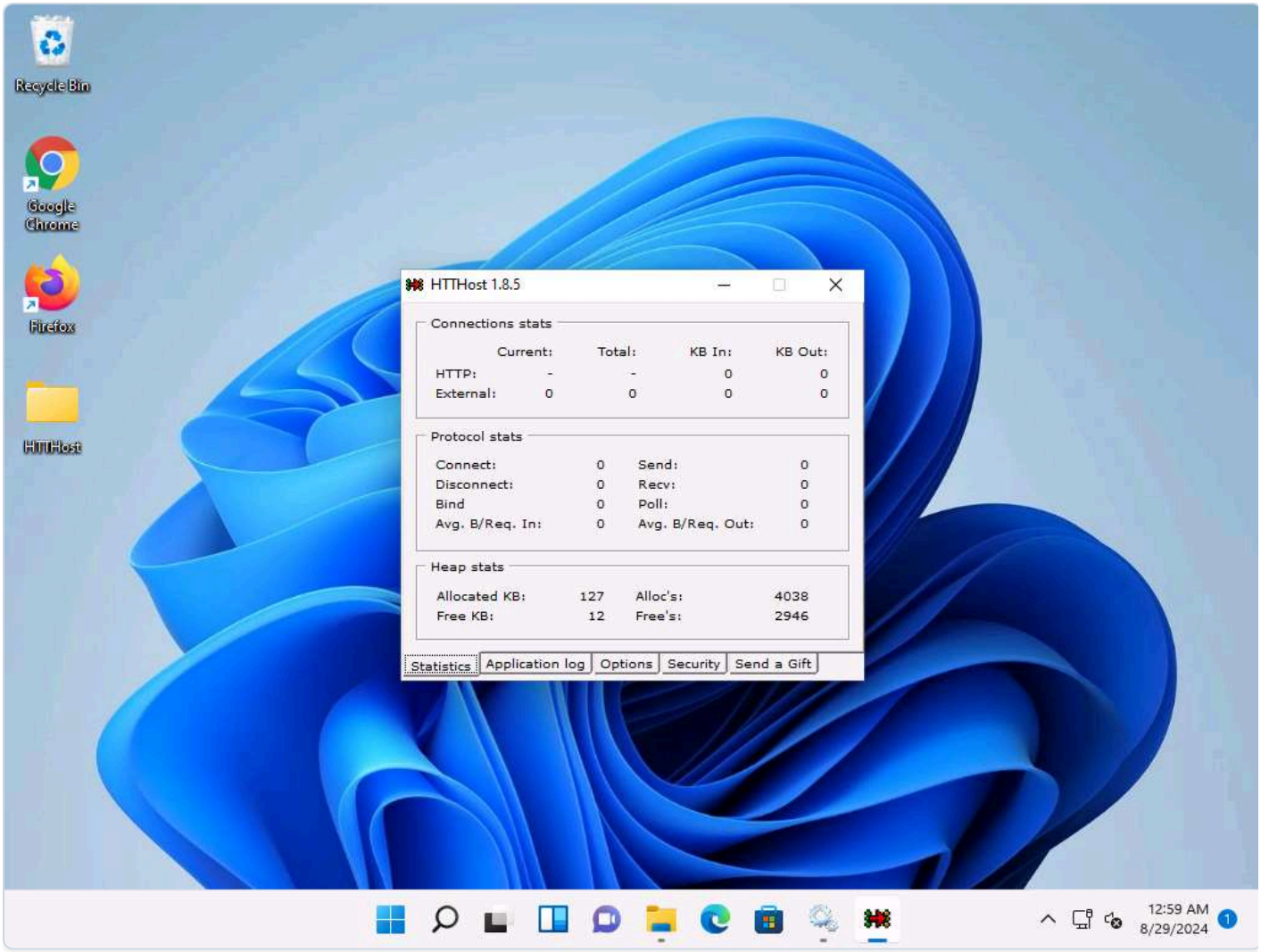


1. Locate **IIS Admin Service**, right-click, and select **Stop**. Minimize the Services console.

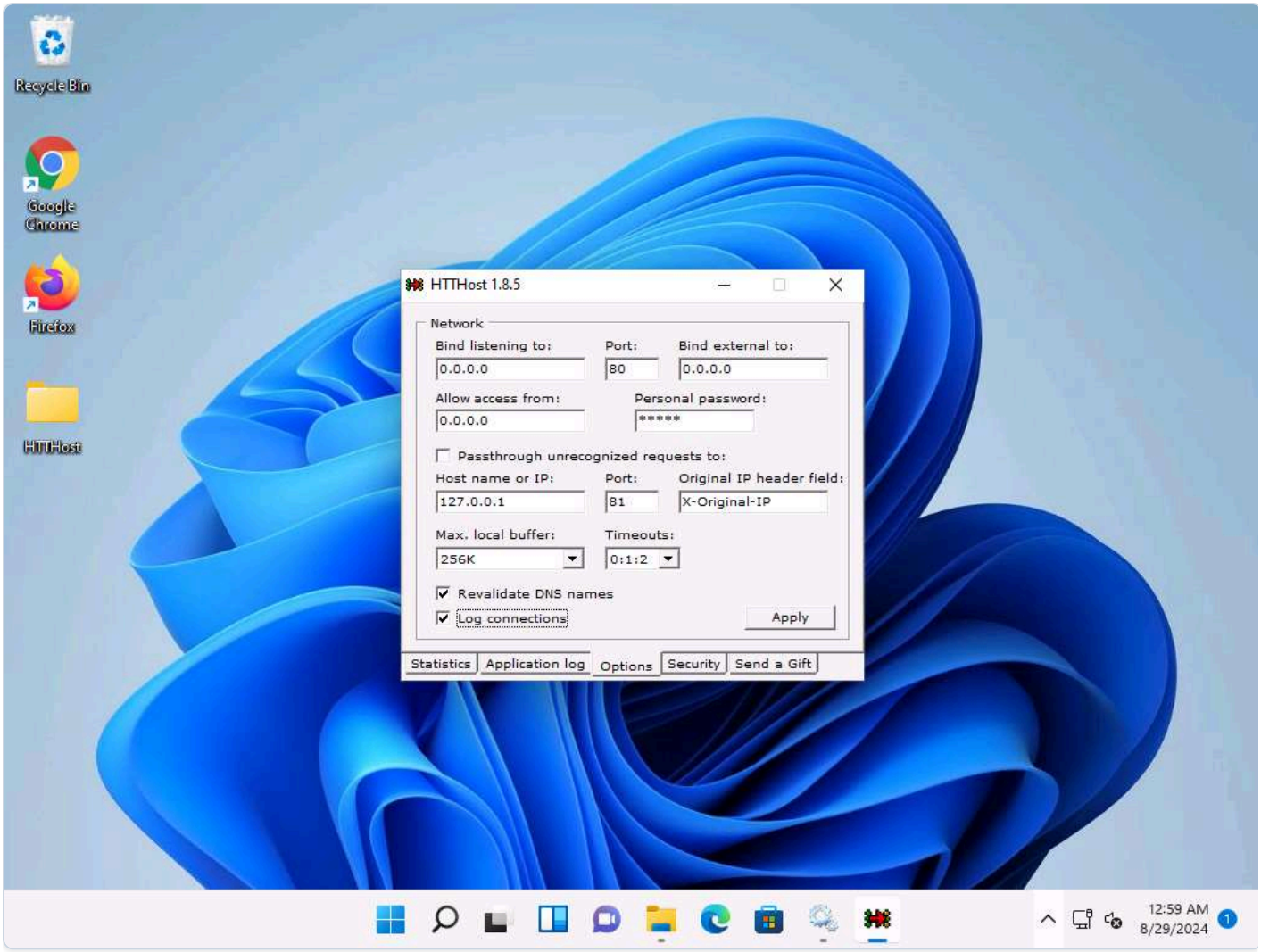


1. Open the `HTTHost` folder on the Desktop and execute **htthost.exe**. Click **Run** if prompted.

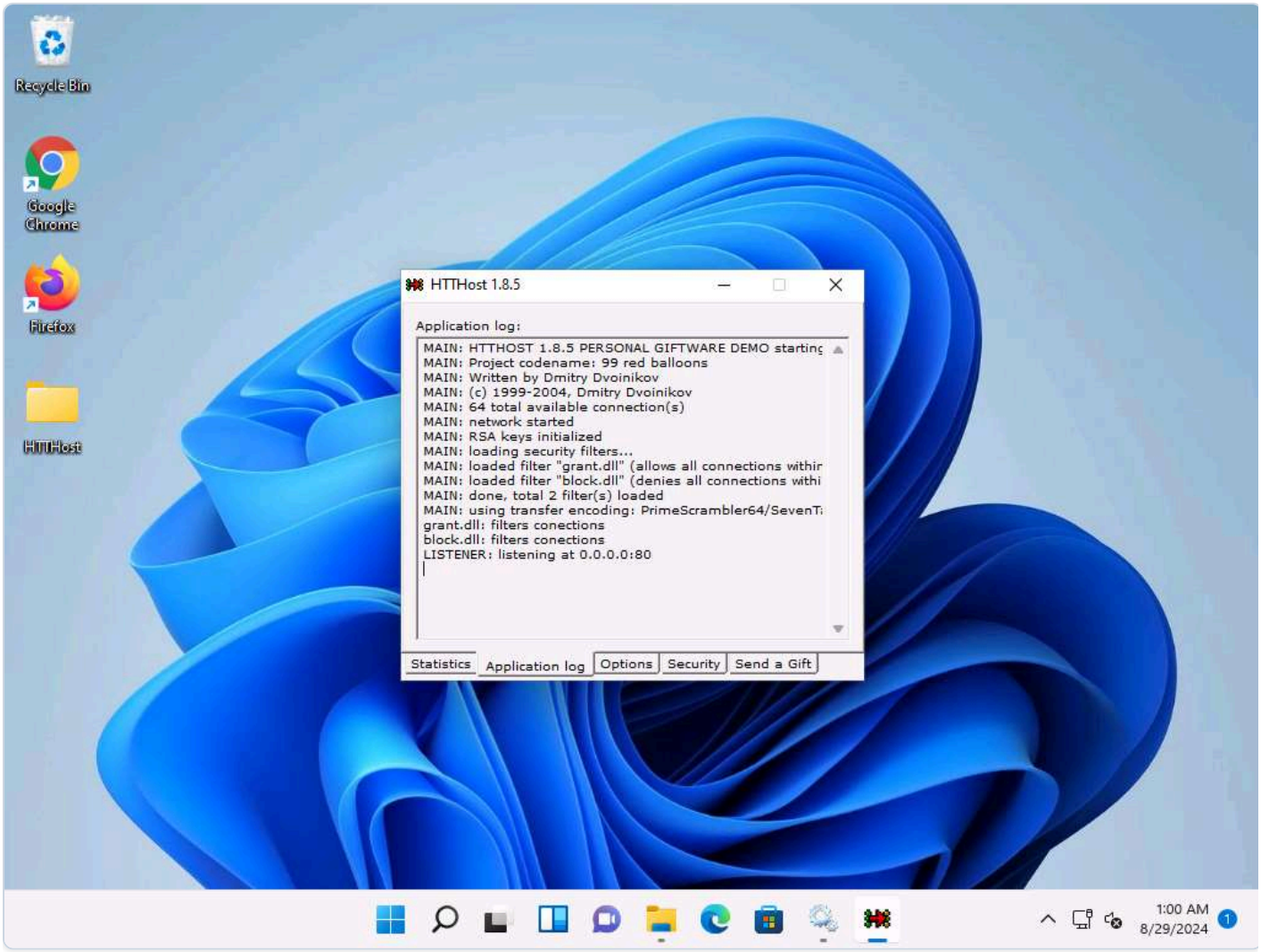

HTTHost
New
Sort
View
HTTHost
Search HTTHost
Quick access
OneDrive - Personal
This PC
Network
Name
Date modified
Type
Size
LOGS 9/7/2004 10:39 AM File folder
block.dll 12/8/2002 11:00 PM Application exten... 40 KB
block.dll.sig 12/8/2002 11:00 PM SIG File 1 KB
eula.txt 9/7/2004 10:30 AM Text Document 7 KB
filters.cfg 12/8/2002 11:00 PM CFG File 1 KB
grant.dll 12/8/2002 11:00 PM Application exten... 40 KB
grant.dll.sig 12/8/2002 11:00 PM SIG File 1 KB
htthost.exe 9/7/2004 10:26 AM Application 444 KB
htthost.exe.sig 9/7/2004 10:29 AM SIG File 1 KB
htthost.ini 9/7/2004 10:39 AM Configuration sett... 1 KB
htthost.pri 12/8/2002 11:00 PM PRI File 1 KB
htthost.pub 12/8/2002 11:00 PM PUB File 1 KB
htthostc.exe 9/7/2004 10:26 AM Application 373 KB
htthostc.exe.sig 9/7/2004 10:29 AM SIG File 1 KB
readme.txt 9/7/2004 9:25 AM Text Document 10 KB
rkeyproc.dll 12/8/2002 11:00 PM Application exten... 19 KB
rkeyproc.dll.sig 12/8/2002 11:00 PM SIG File 1 KB
19 items 1 item selected 444 KB
Thursday, August 29, 2024
12:58 AM
8/29/2024



1. Select the **Options** tab (**options**). Set **Personal Password** to **magic**, check **Revalidate DNS names** and **Log Connections**, then click **Apply**.



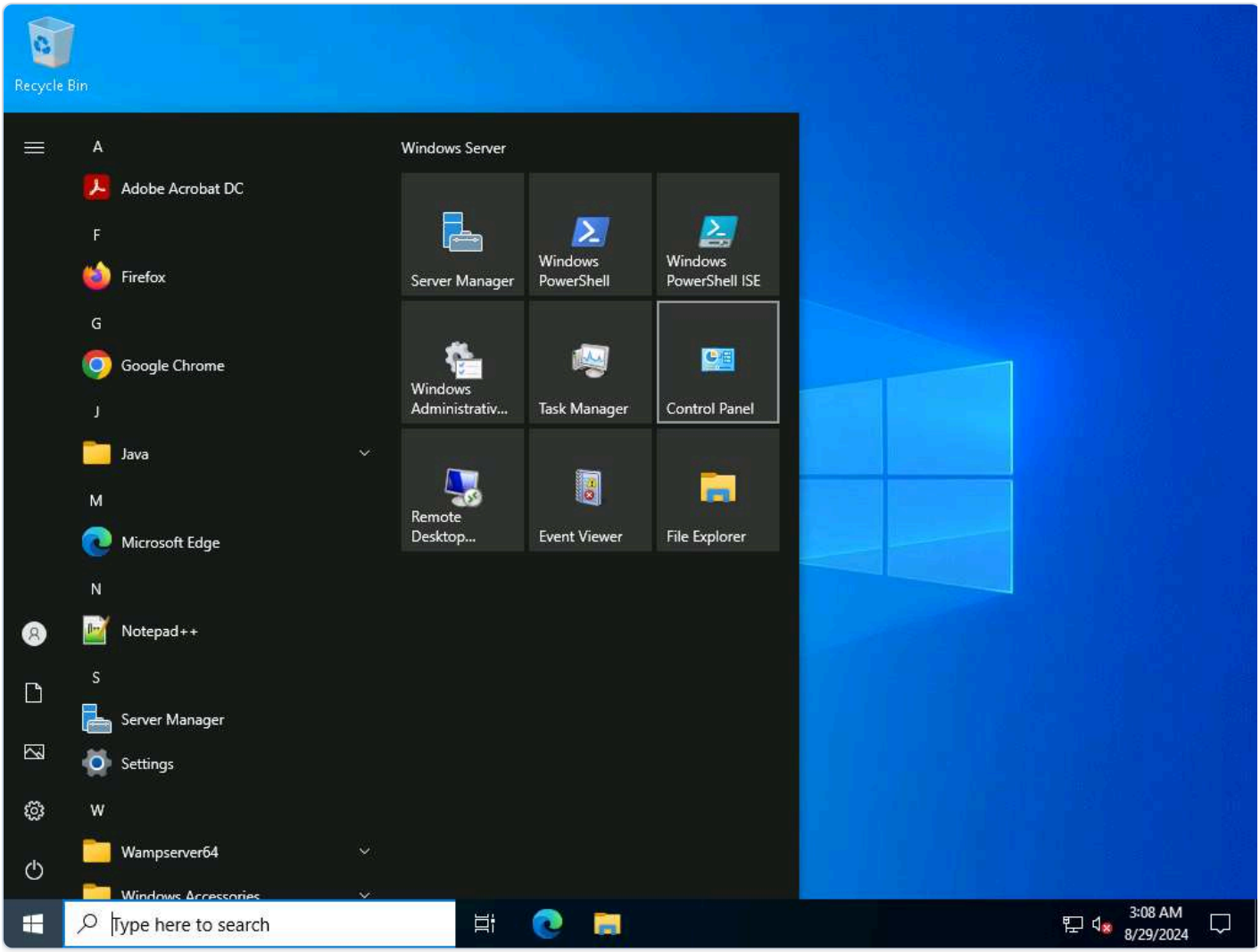
1. Verify in the Application Log pane that the listener status confirms: `Listener: listening at 0.0.0.0:80`. Keep HTTHost running.

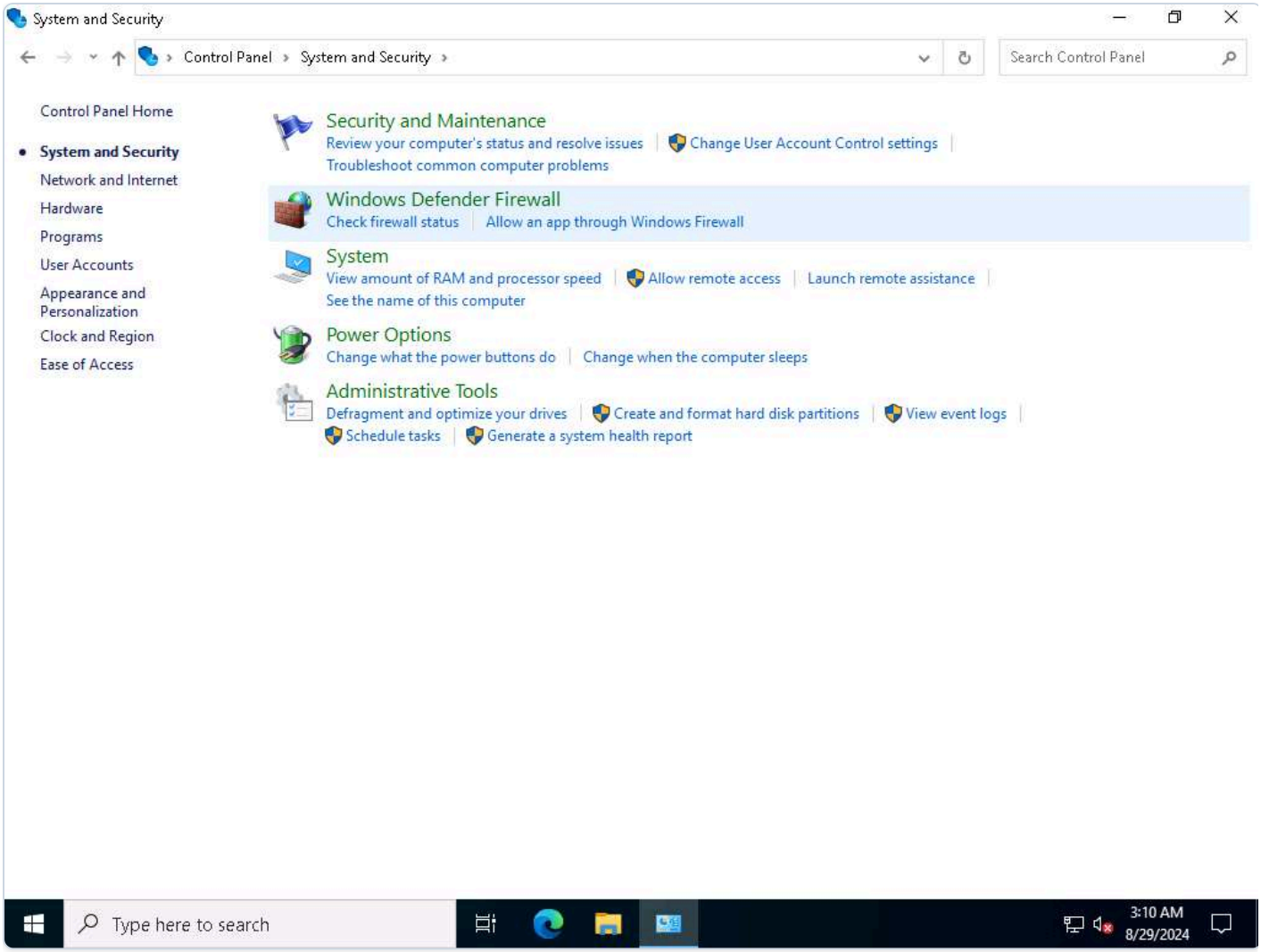


Task 2: Configure Outbound Firewall Block on Windows Server 2022

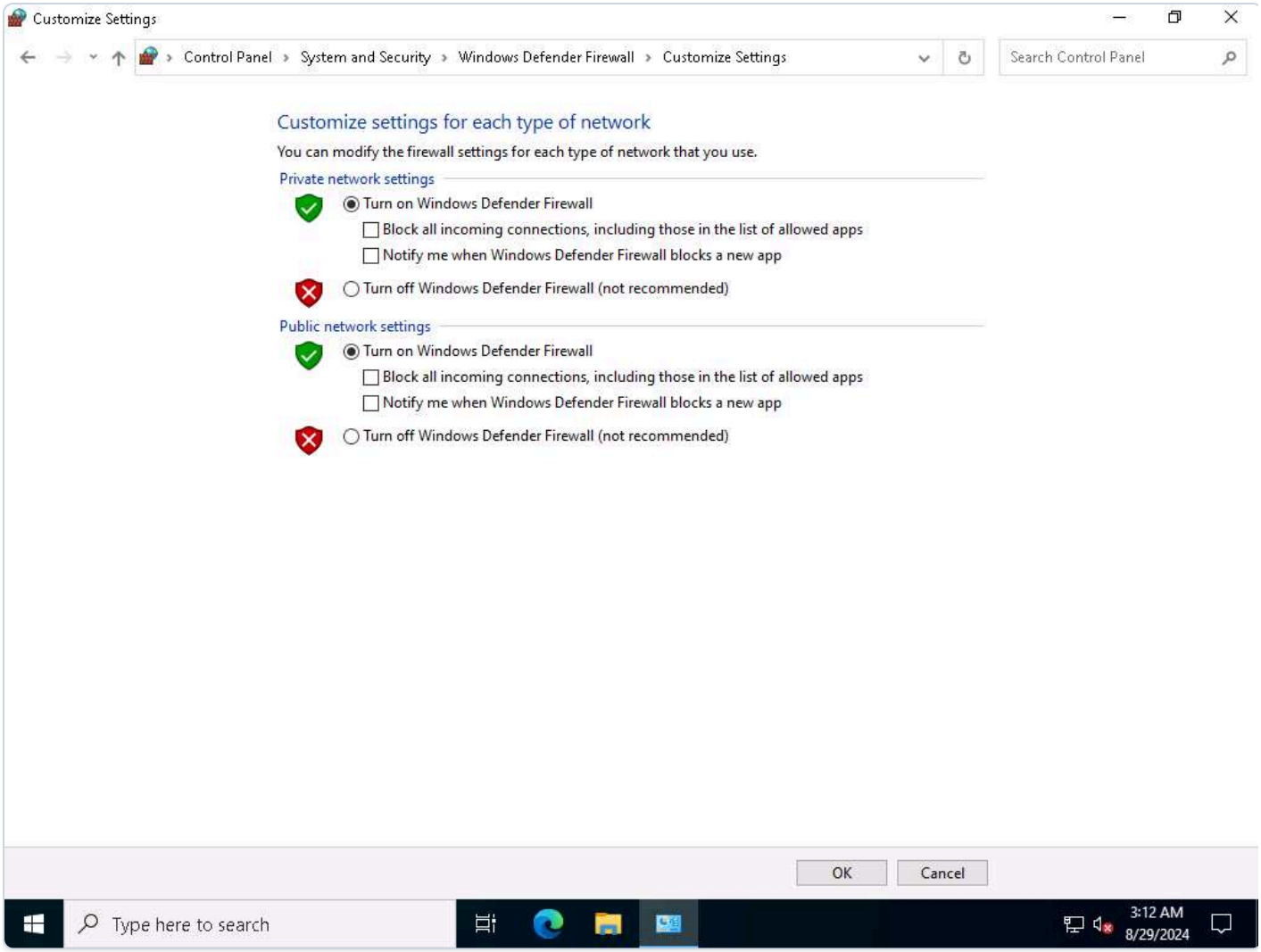
In this task, we log into Windows Server 2022, enable Windows Defender Firewall, create an outbound blocking rule targeting remote port 21, and confirm that direct FTP connections to Sales Department (**172.19.19.17**) are blocked.

1. Switch to **Windows Server 2022** and log in with password **Pa$$w0rd**.
2. Open **Control Panel** $
   ightarrow$ **System and Security** $
   ightarrow$ **Windows Defender Firewall**.

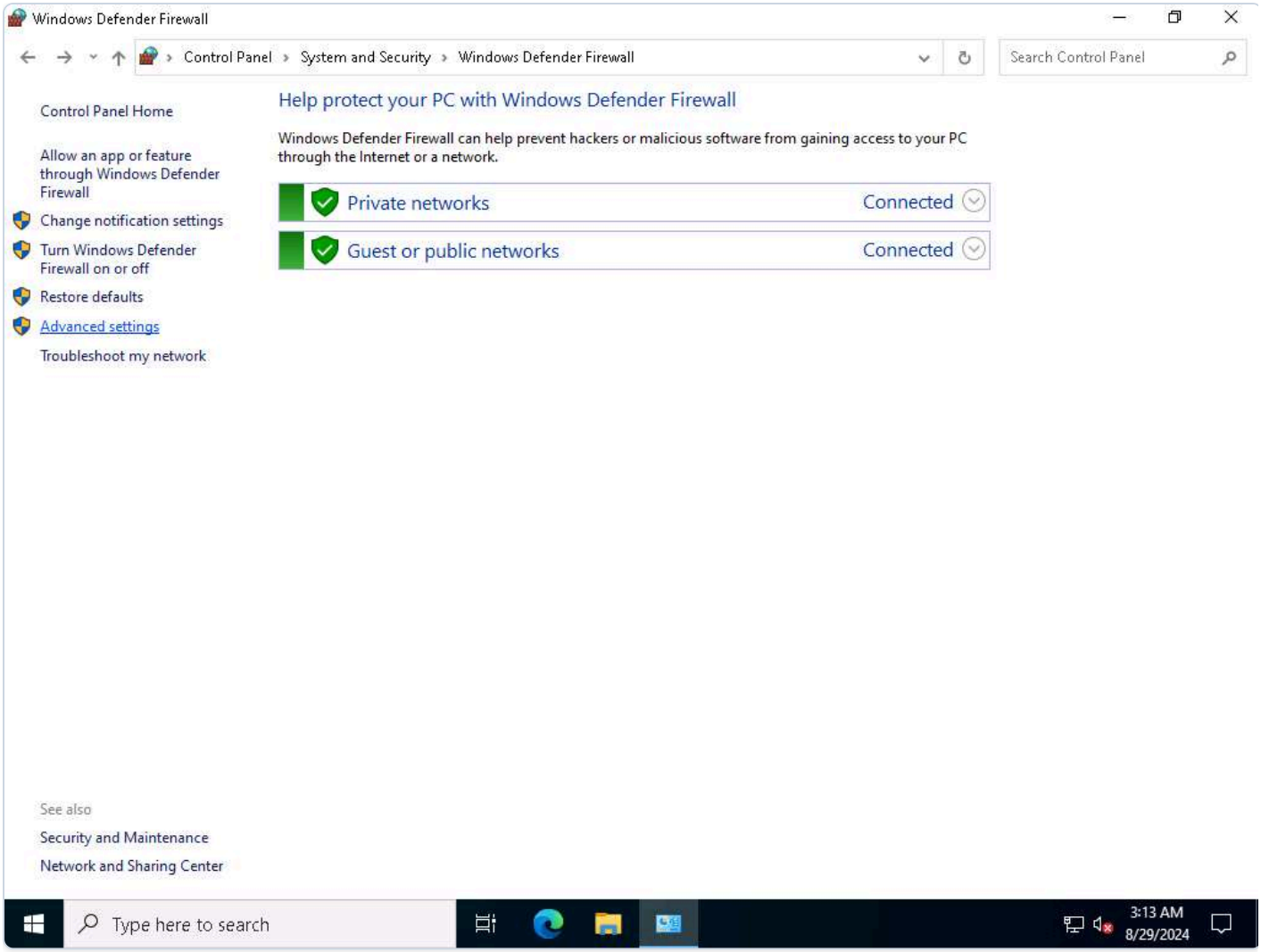

Recycle Bin
A
Adobe Acrobat DC
F
Firefox
G
Google Chrome
J
Java
M
Microsoft Edge
N
Notepad++
S
Server Manager
Settings
W
Wampserver64
Windows Server
Server Manager
Windows PowerShell
Windows PowerShell ISE
Windows Administrativ...
Task Manager
Control Panel
Remote Desktop...
Event Viewer
File Explorer
Type here to search
3:08 AM
8/29/2024



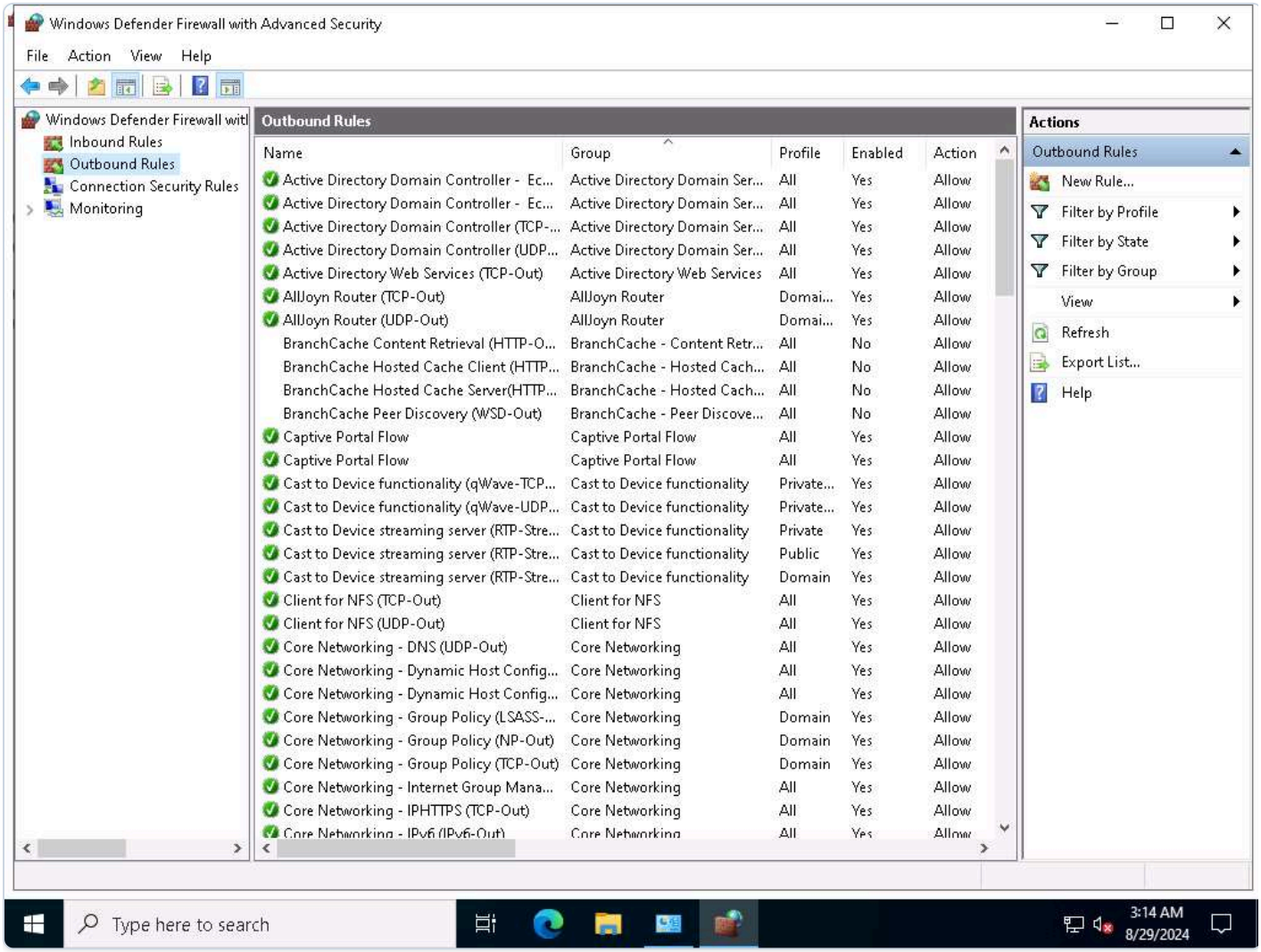
1. Click **Turn Windows Firewall on or off**. Turn ON Windows Firewall for both Private and Public network profiles, then click **OK**.



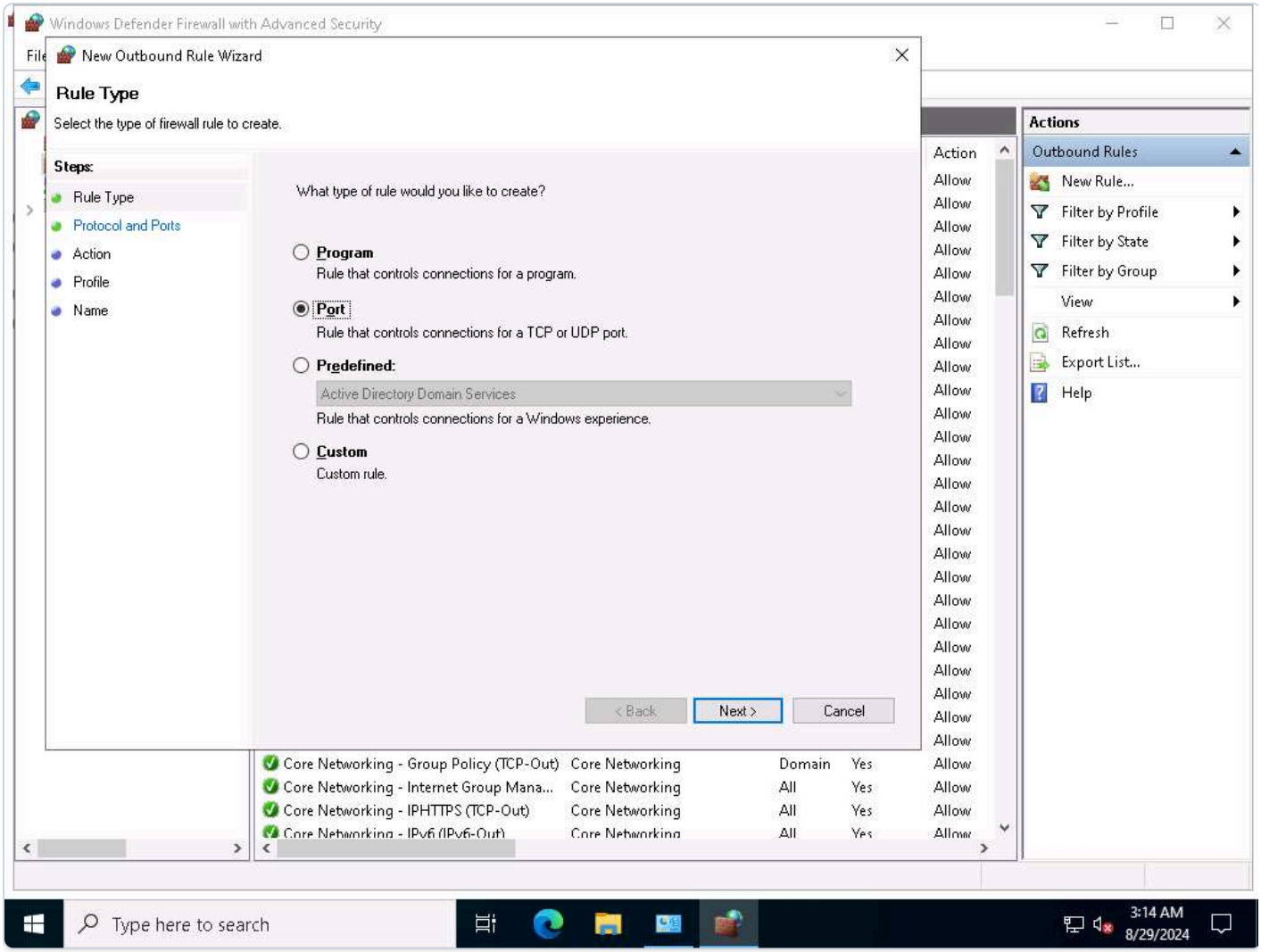
1. Click **Advanced settings** in the left pane.



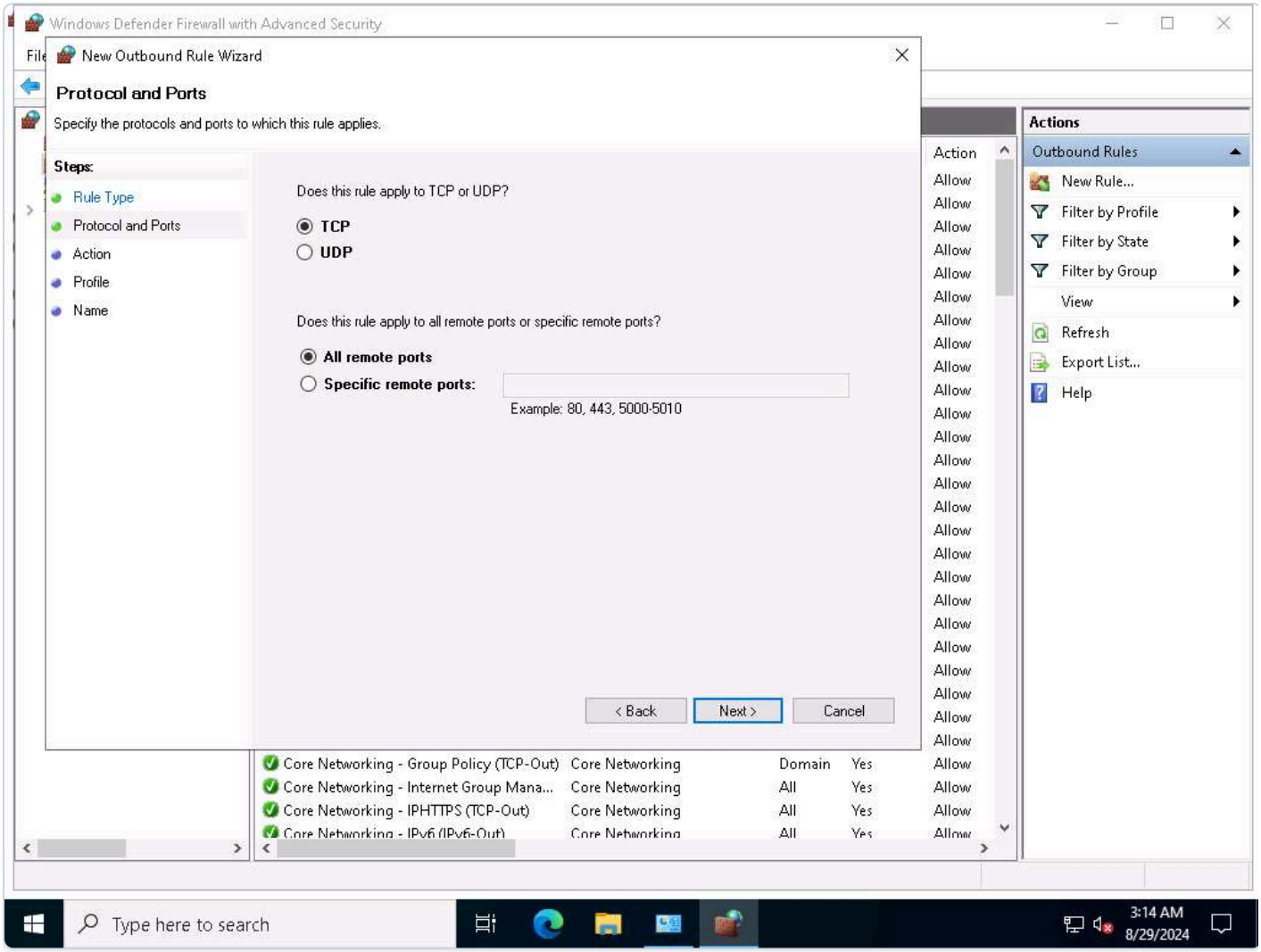
1. Select **Outbound Rules** $
   ightarrow$ click **New Rule...** in the right pane.



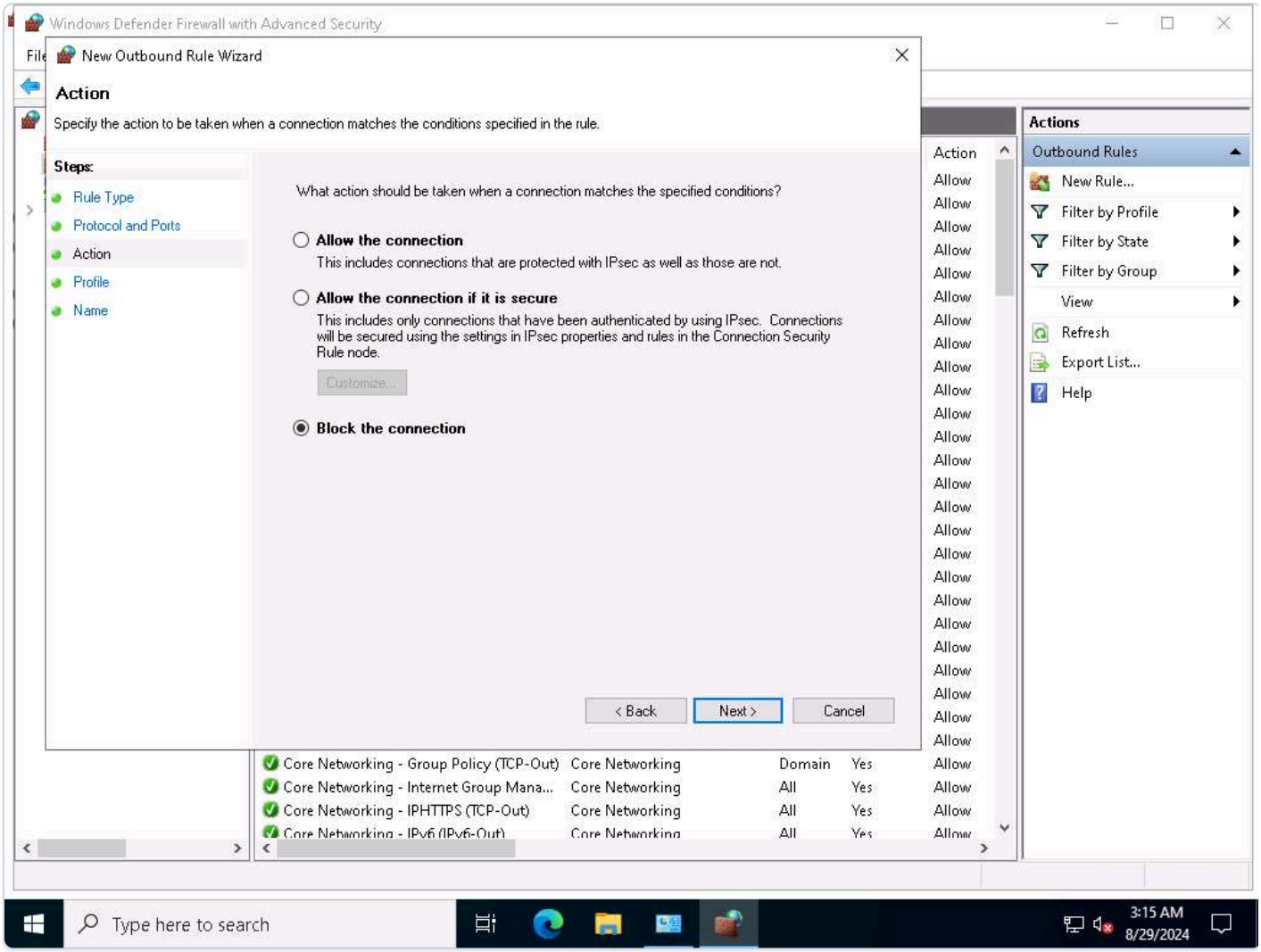
1. Select **Port** as Rule Type, click **Next**.



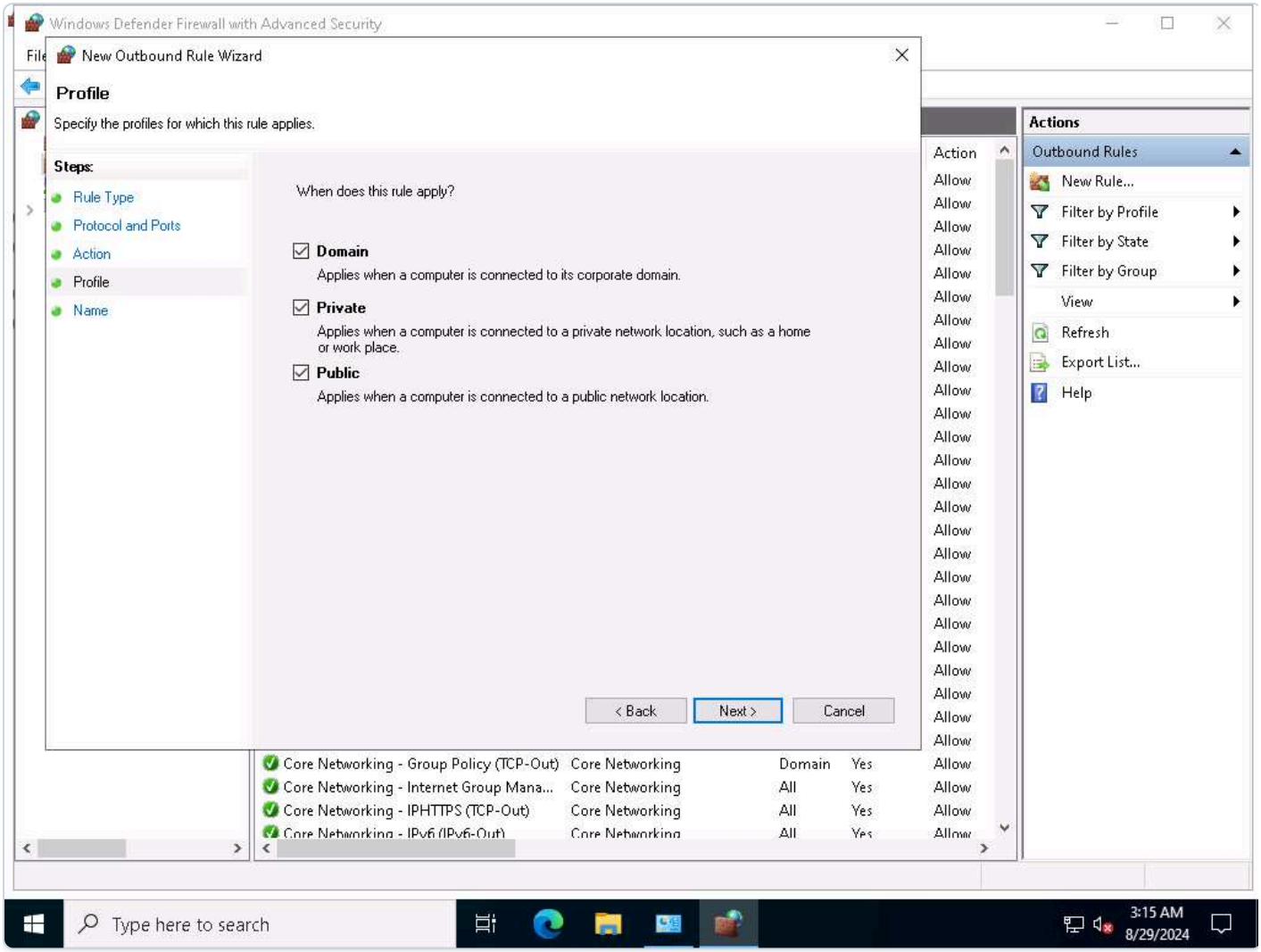
1. Select **All remote ports**, click **Next**.



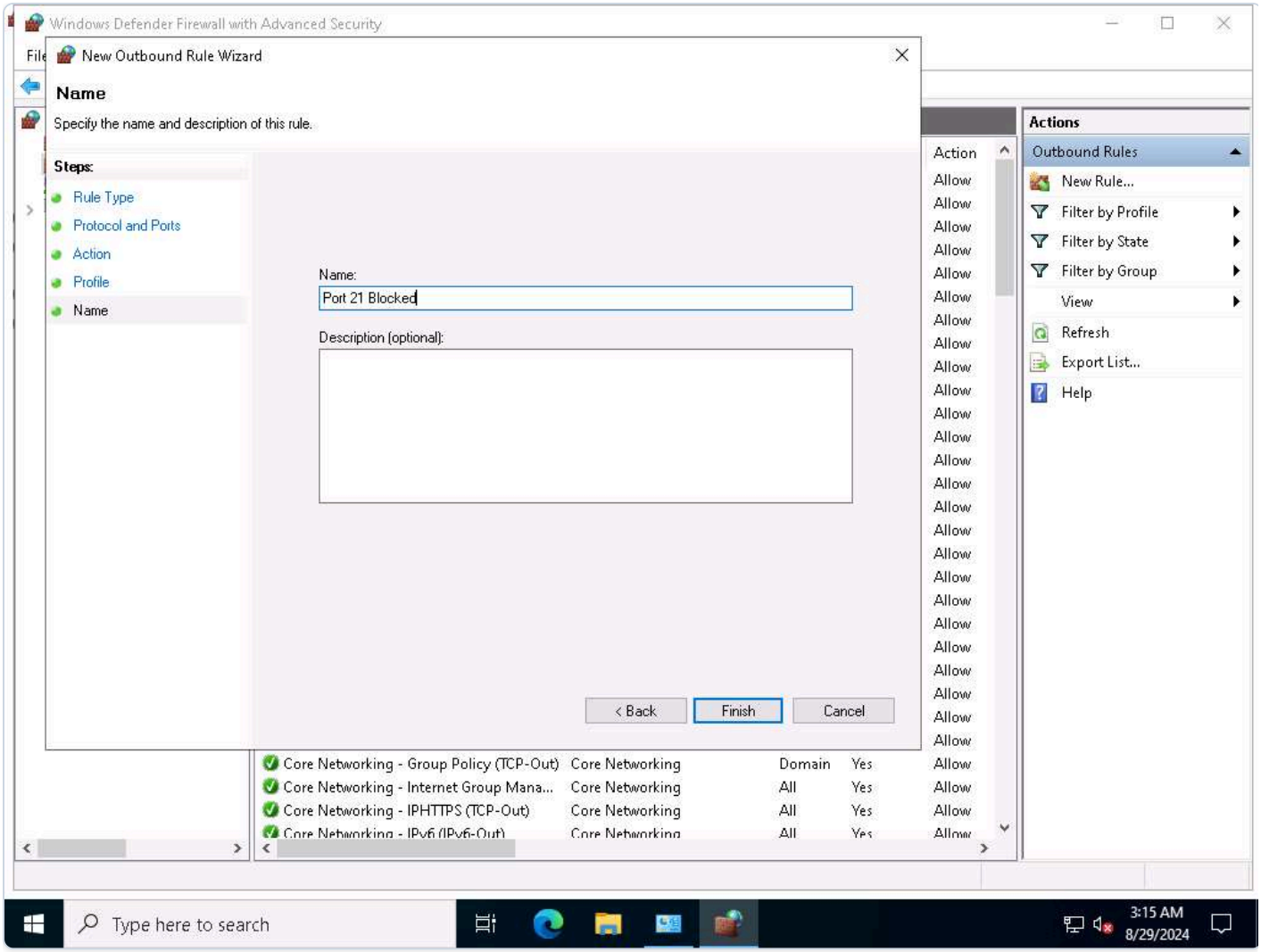
1. Select **Block the connection**, click **Next**.



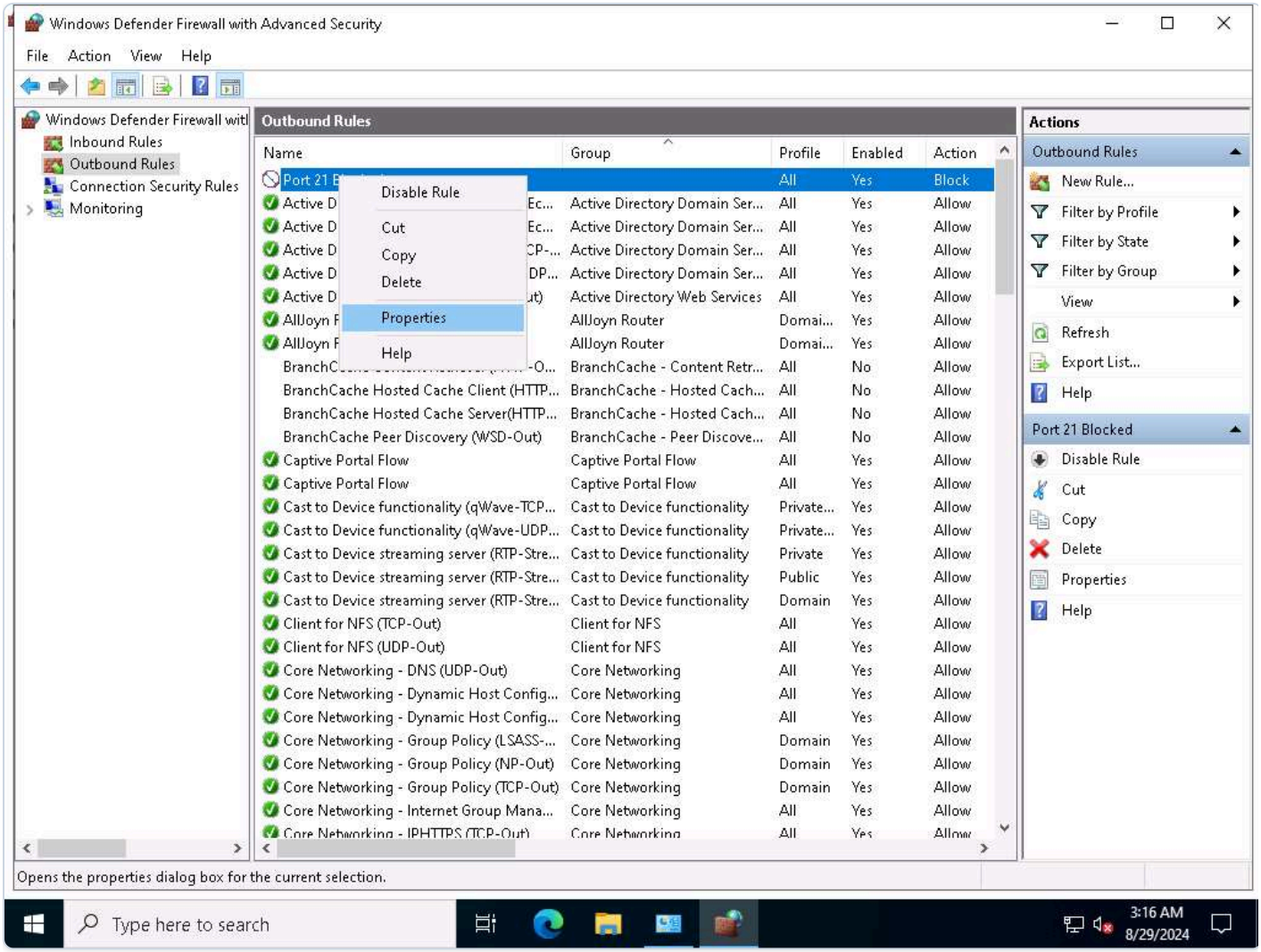
1. Ensure Domain, Private, and Public profiles are checked, click **Next**.



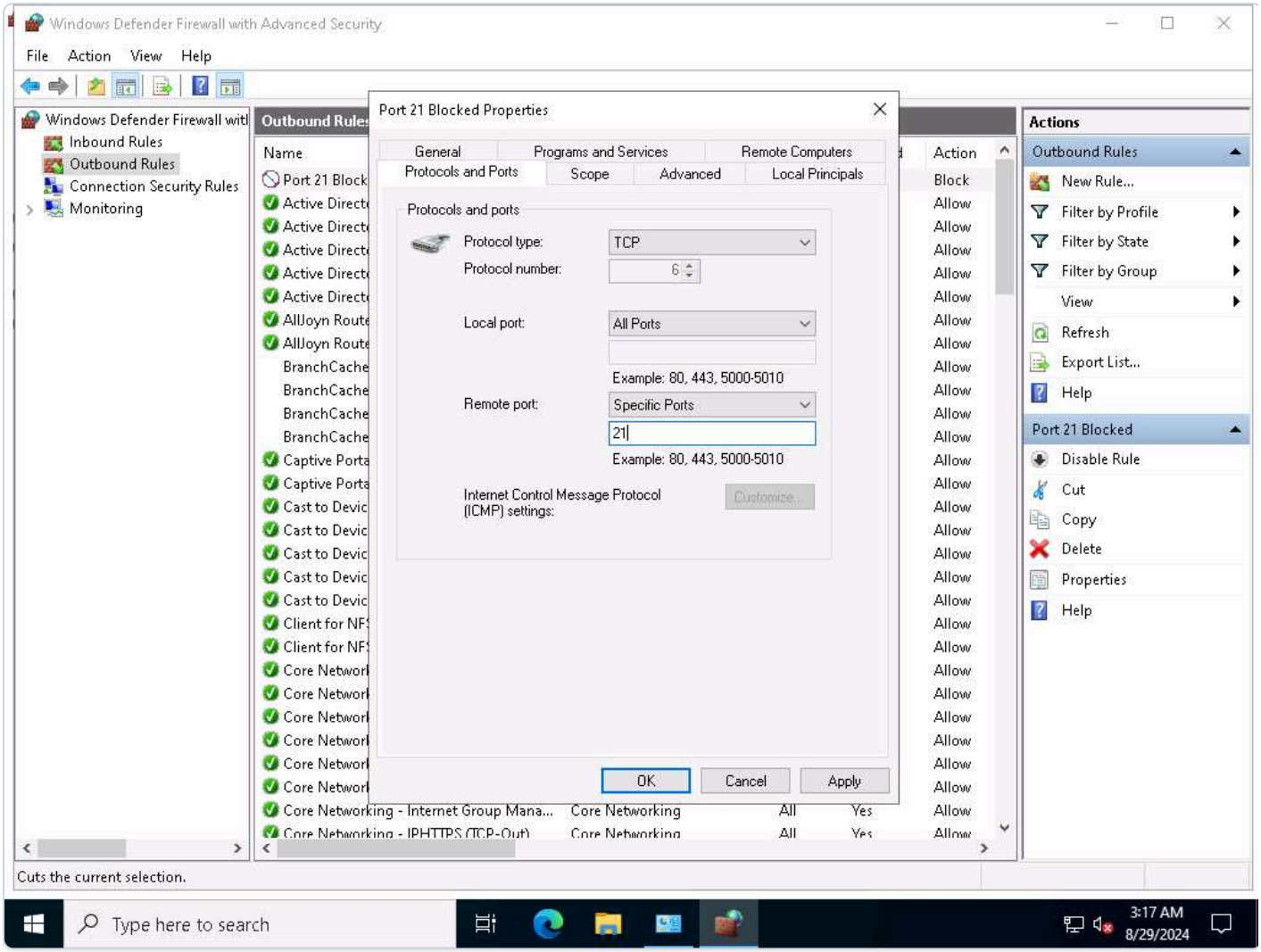
1. Enter Name: **Port 21 Blocked**, click **Finish**.



1. Right-click **Port 21 Blocked** $
   ightarrow$ **Properties** $
   ightarrow$ **Protocols and Ports** tab $
   ightarrow$ Remote Port: **Specific Ports** $
   ightarrow$ specify **21**. Click **Apply** and **OK**.



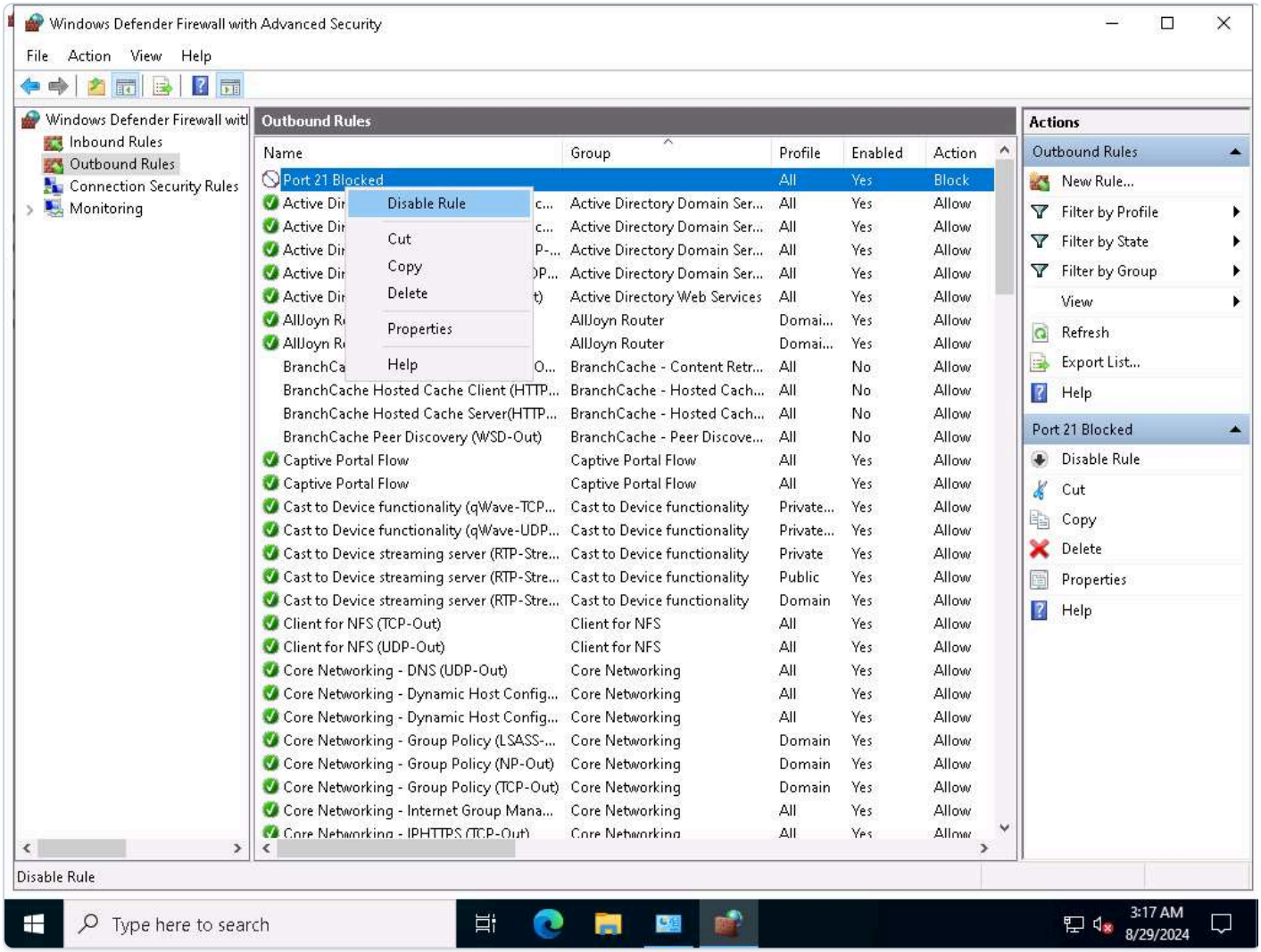
1. Right-click the rule $
   ightarrow$ **Disable Rule**.



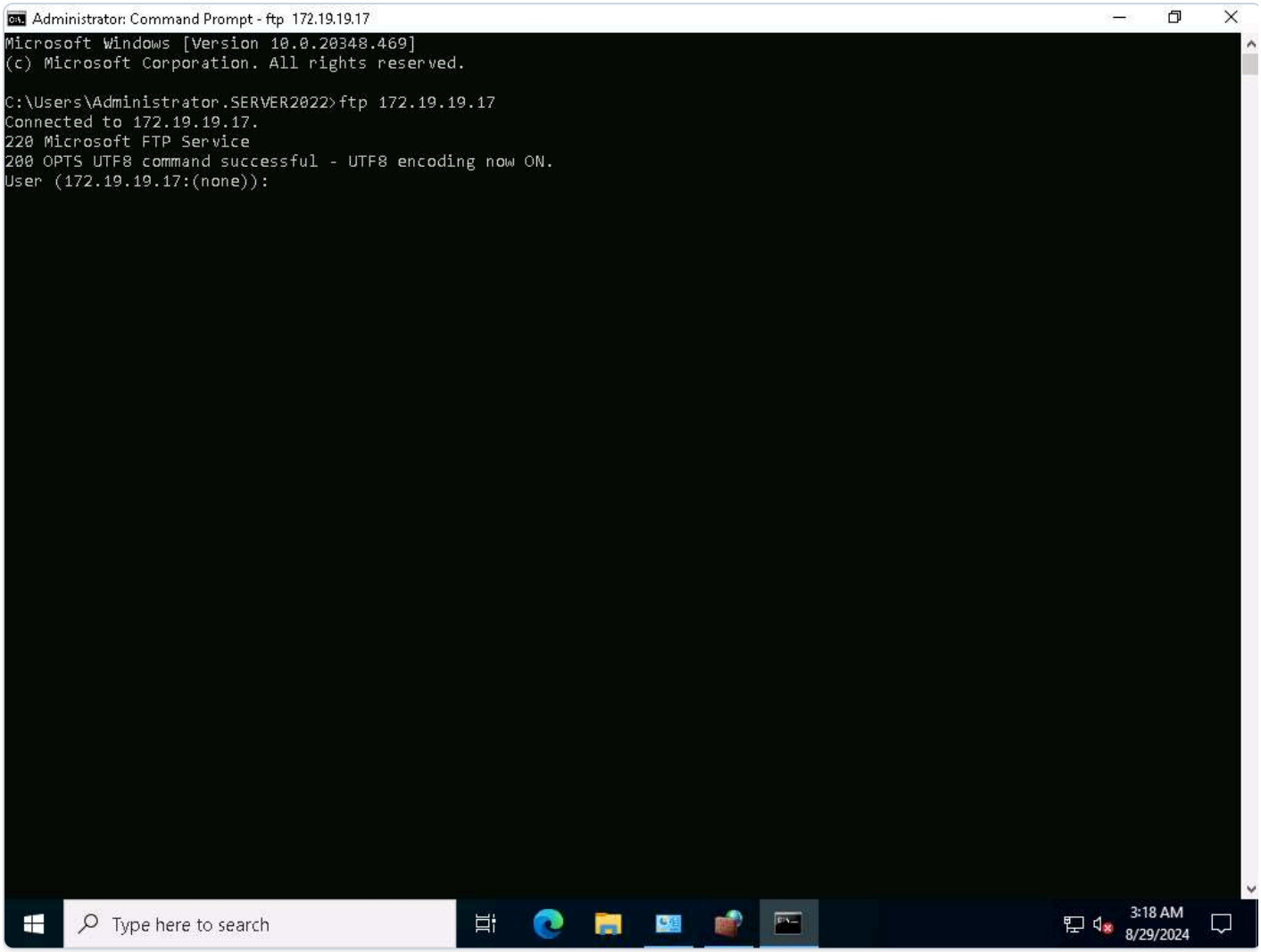
1. Open Command Prompt and test direct FTP connectivity:

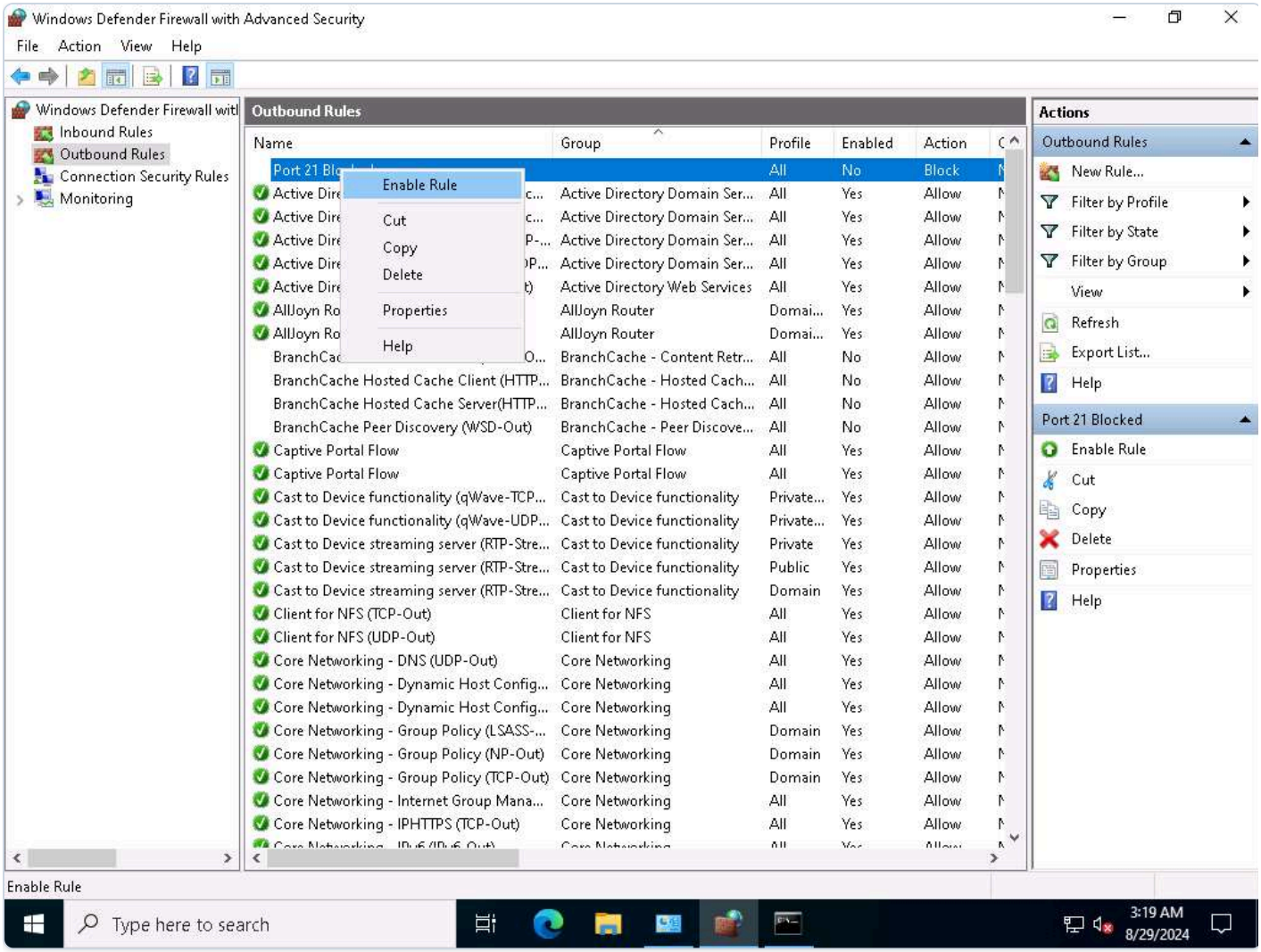
   ftp 172.19.19.17

   (Confirm username prompt appears, then press Ctrl+C).



1. Re-enable the rule (**Enable Rule**) and verify that direct FTP connection attempts are blocked.

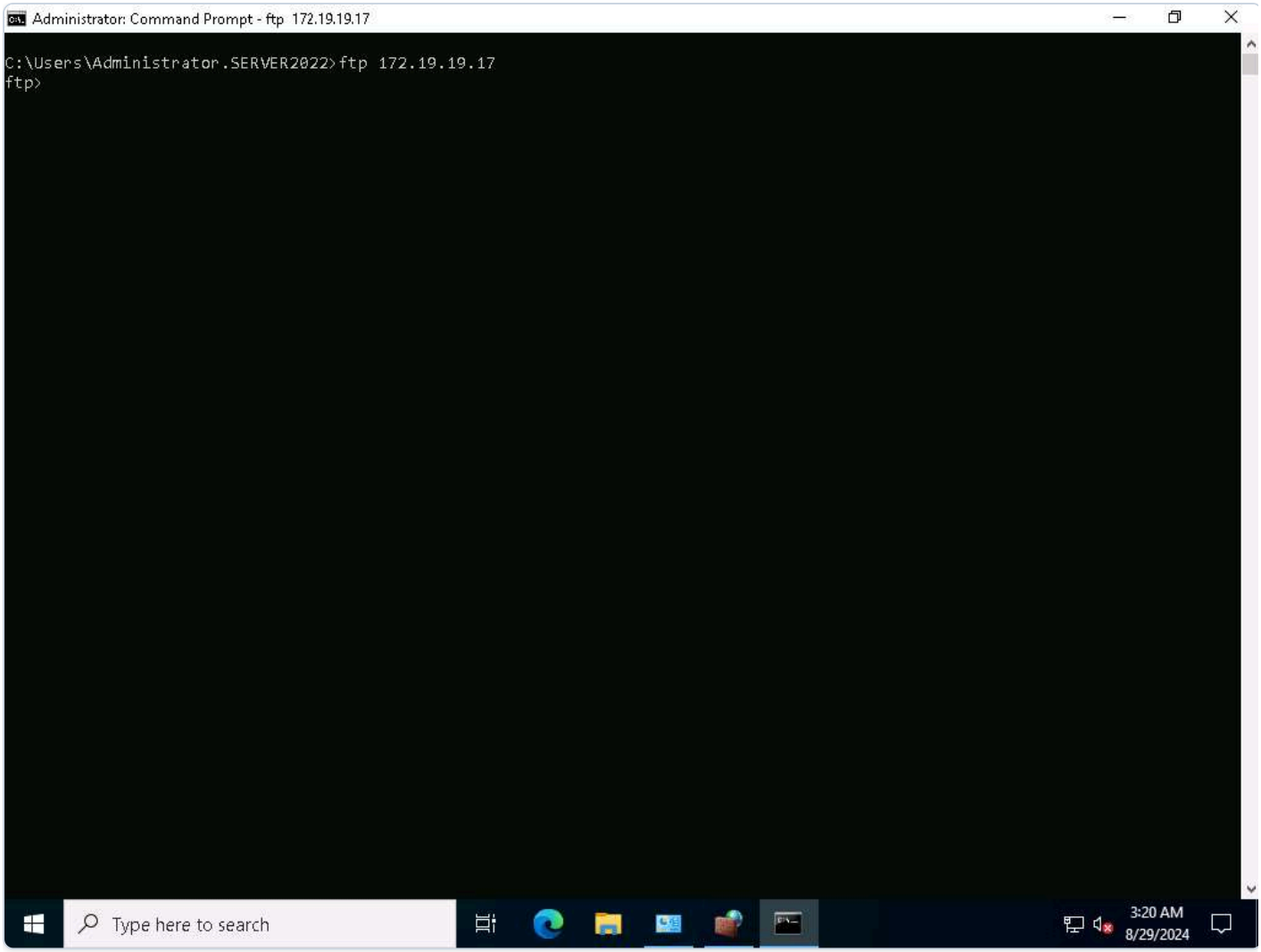

Administrator: Command Prompt - ftp 172.19.19.17
Microsoft Windows [Version 10.0.20348.469]
(c) Microsoft Corporation. All rights reserved.
C:\Users\Administrator.SERVER2022>ftp 172.19.19.17
Connected to 172.19.19.17.
220 Microsoft FTP Service
200 OPTS UTF8 command successful - UTF8 encoding now ON.
User (172.19.19.17:(none)):
Type here to search
3:18 AM
8/29/2024



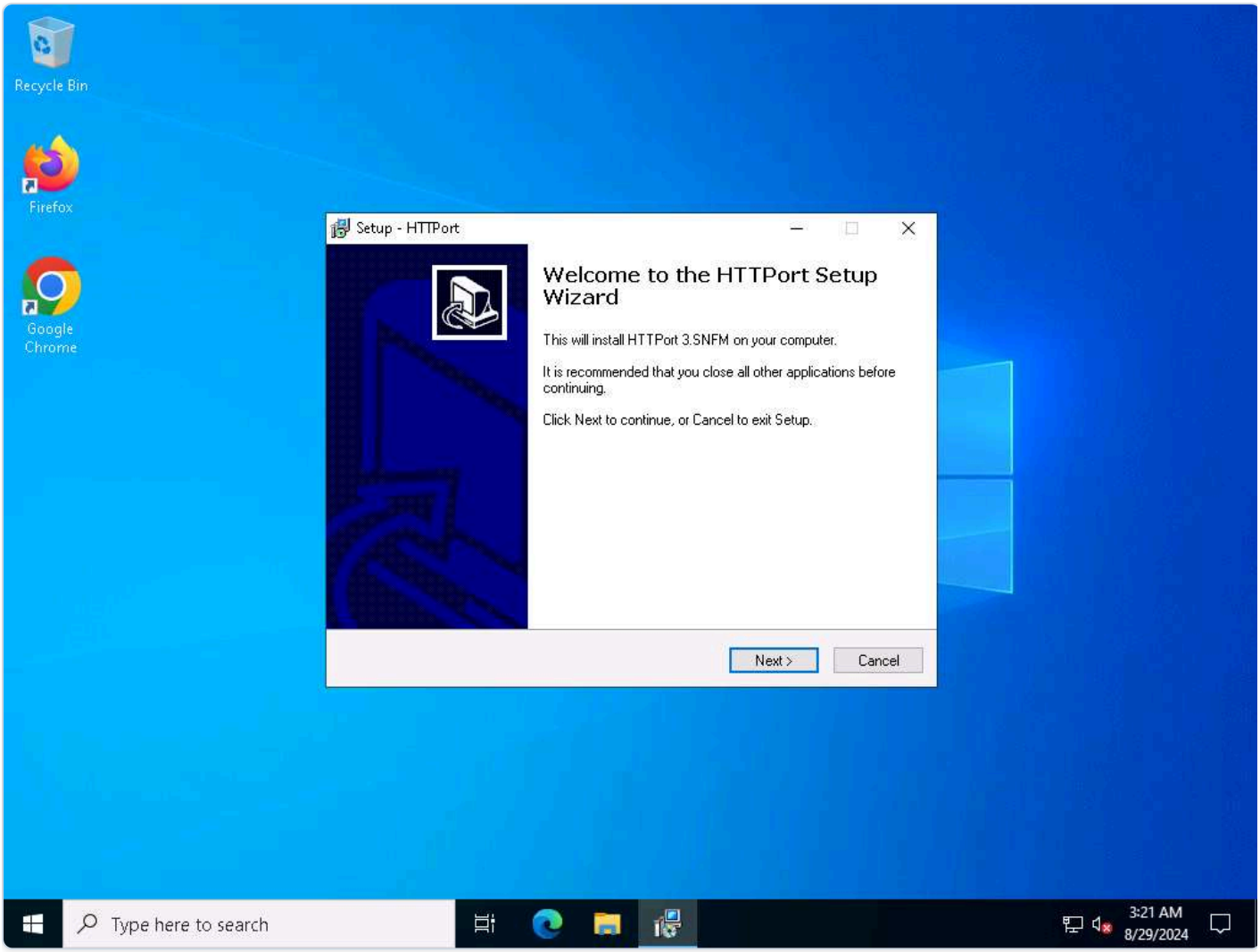
Task 3: Install HTTPort and Execute HTTP Tunneling

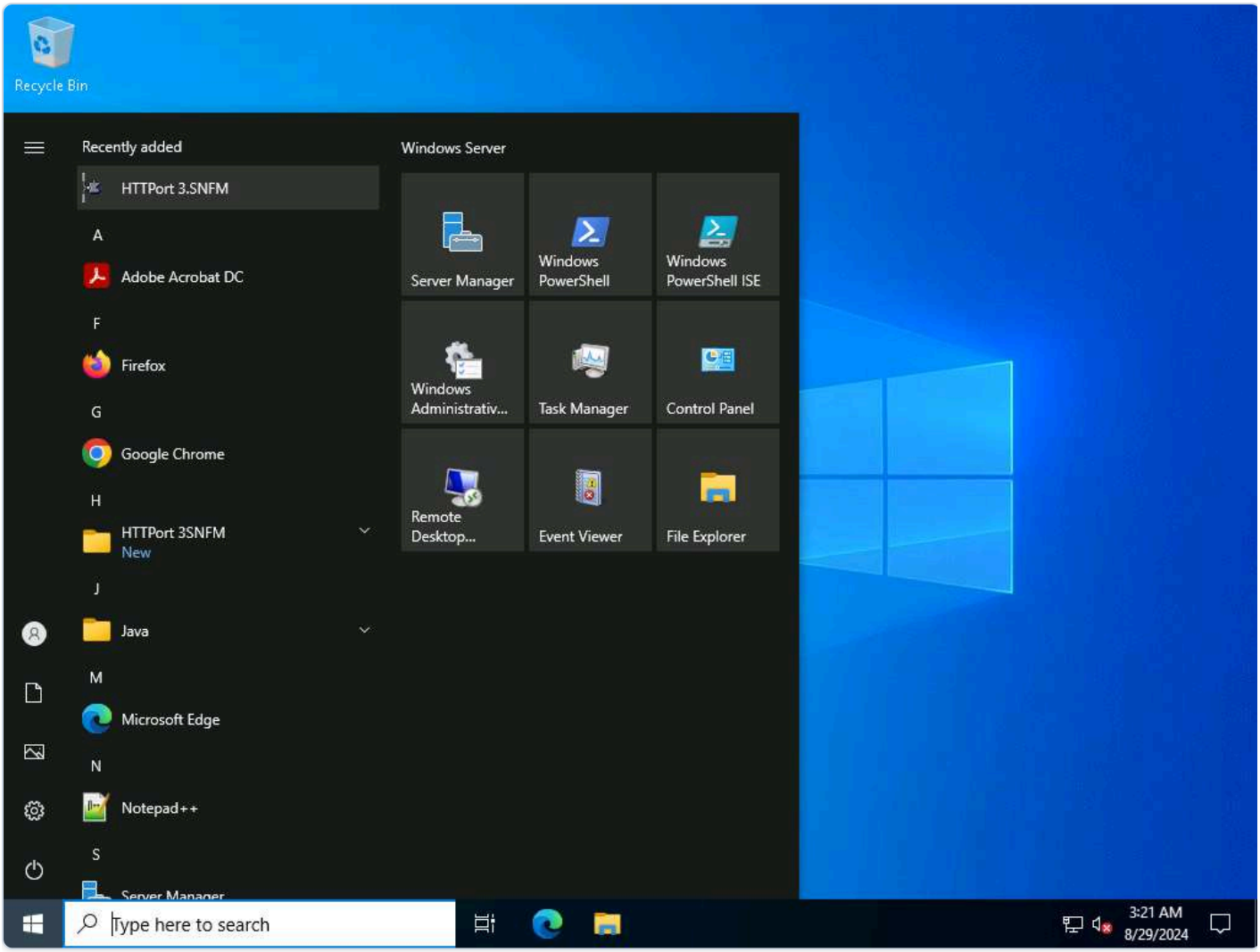
In this task, we install HTTPort on Windows Server 2022, configure proxy and remote host settings targeting HTTHost (**172.19.19.9:80**), map local port 21 to remote Sales Department (**172.19.19.17:21**), launch the tunnel, and perform authenticated FTP directory operations via **ftp 127.0.0.1**.

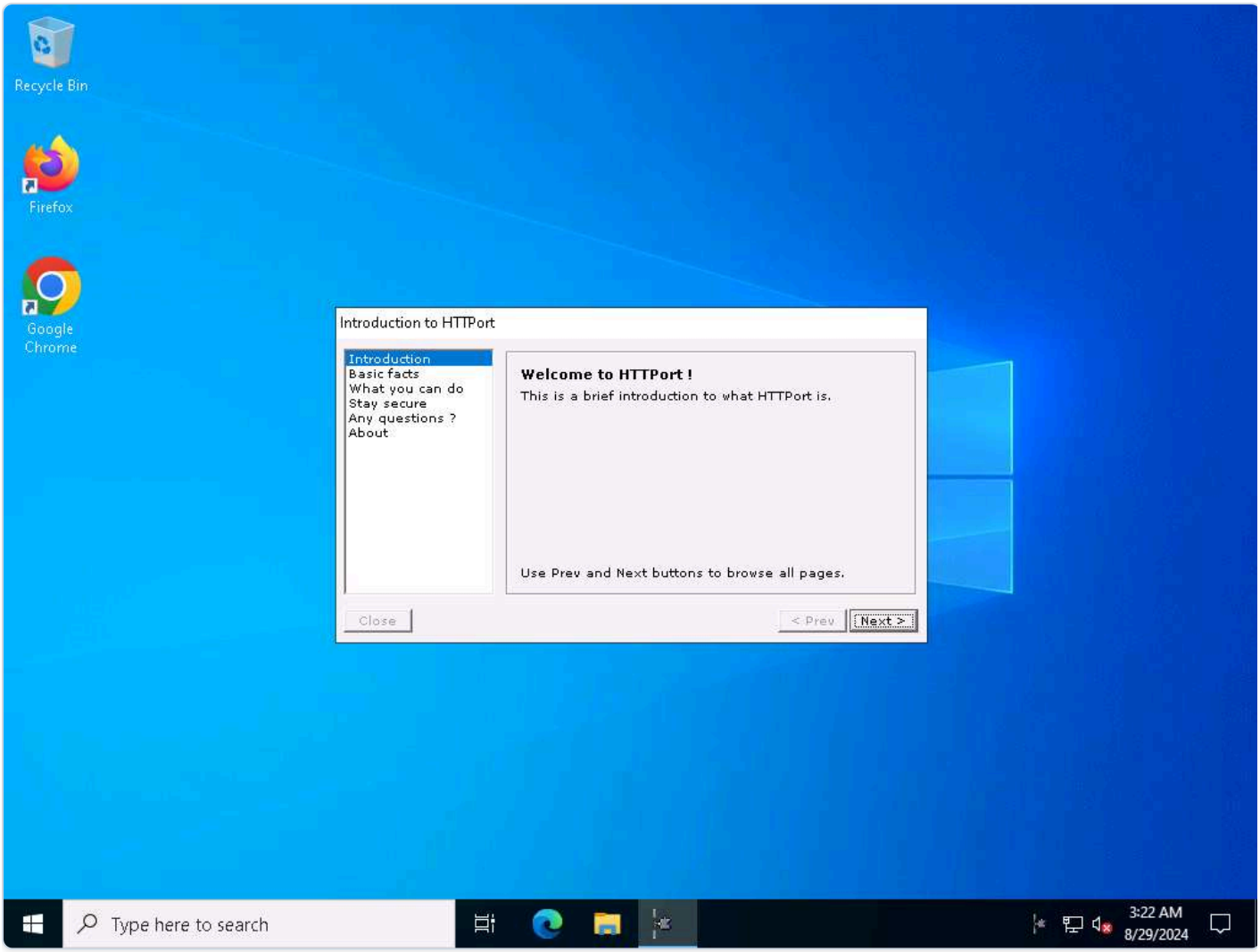
1. Run `C:\CPENT-Tools\CPENT Module 07 Perimeter Defence Evasion Techniques\HTTP Tunneling\httport3snfm.exe` and complete installation.



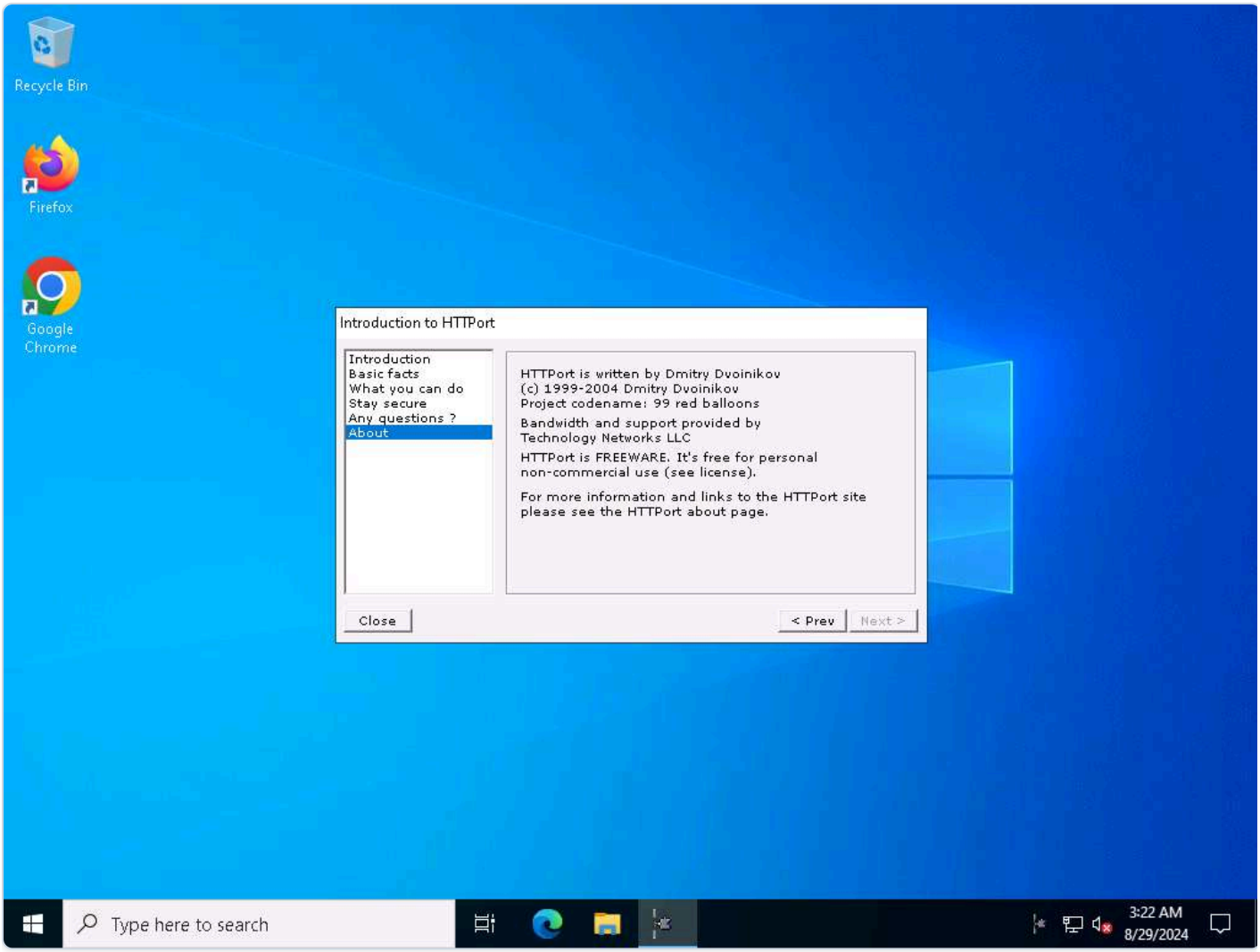
1. Launch **HTTPort 3.SNFM** from the Start menu and complete the initial introduction wizard.


Recycle Bin
Firefox
Google Chrome
Setup - HTTPort
Welcome to the HTTPort Setup Wizard
This will install HTTPort 3.SNFM on your computer.
It is recommended that you close all other applications before continuing.
Click Next to continue, or Cancel to exit Setup.
Next >
Cancel
Type here to search
3:21 AM
8/29/2024

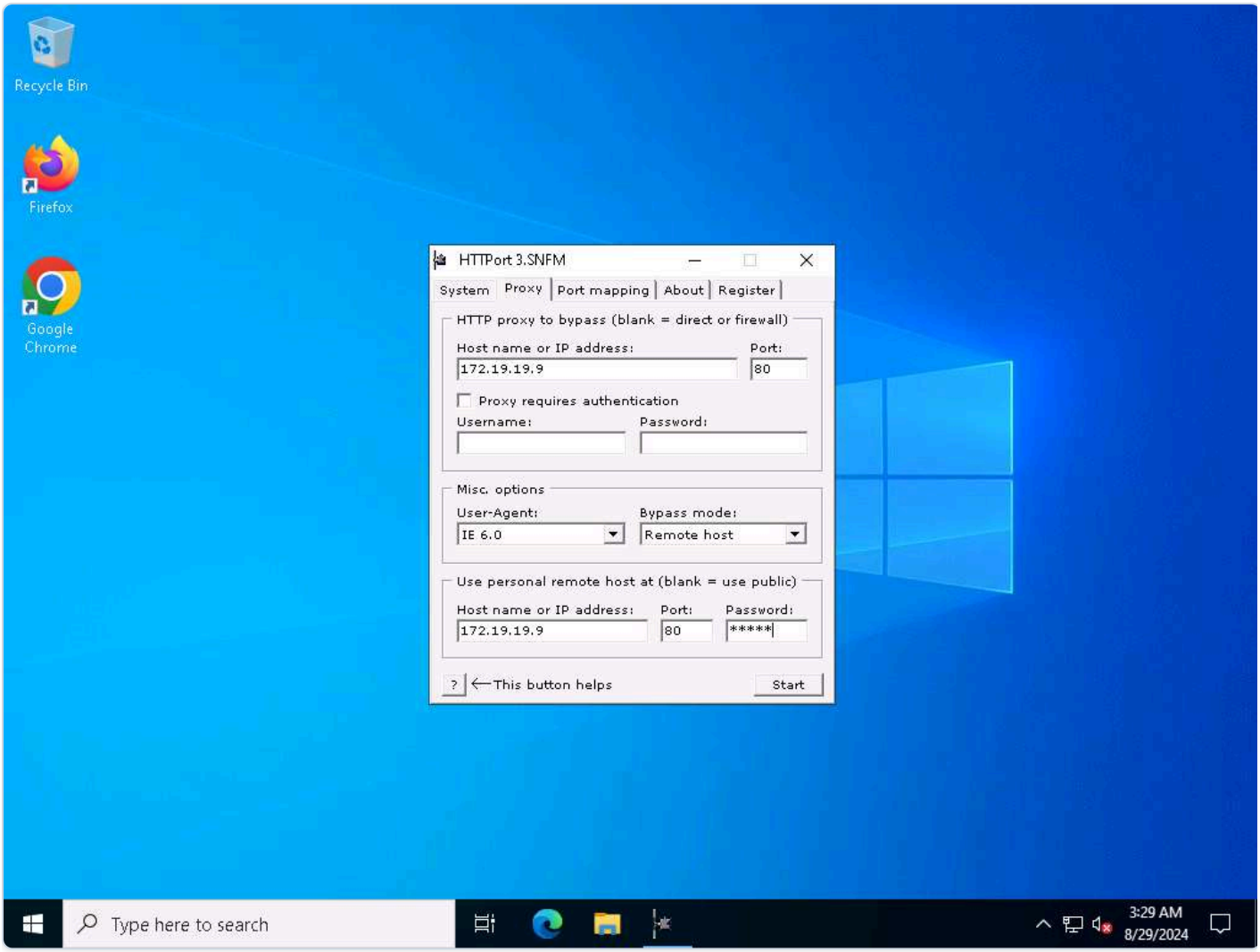

Recycle Bin
Recently added
HTTPort 3.SNFM
A
Adobe Acrobat DC
F
Firefox
G
Google Chrome
H
HTTPort 3SNFM
New
J
Java
M
Microsoft Edge
N
Notepad++
S
Server Manager
Windows Server
Server Manager
Windows PowerShell
Windows PowerShell ISE
Windows Administrativ...
Task Manager
Control Panel
Remote Desktop...
Event Viewer
File Explorer
Type here to search
3:21 AM
8/29/2024



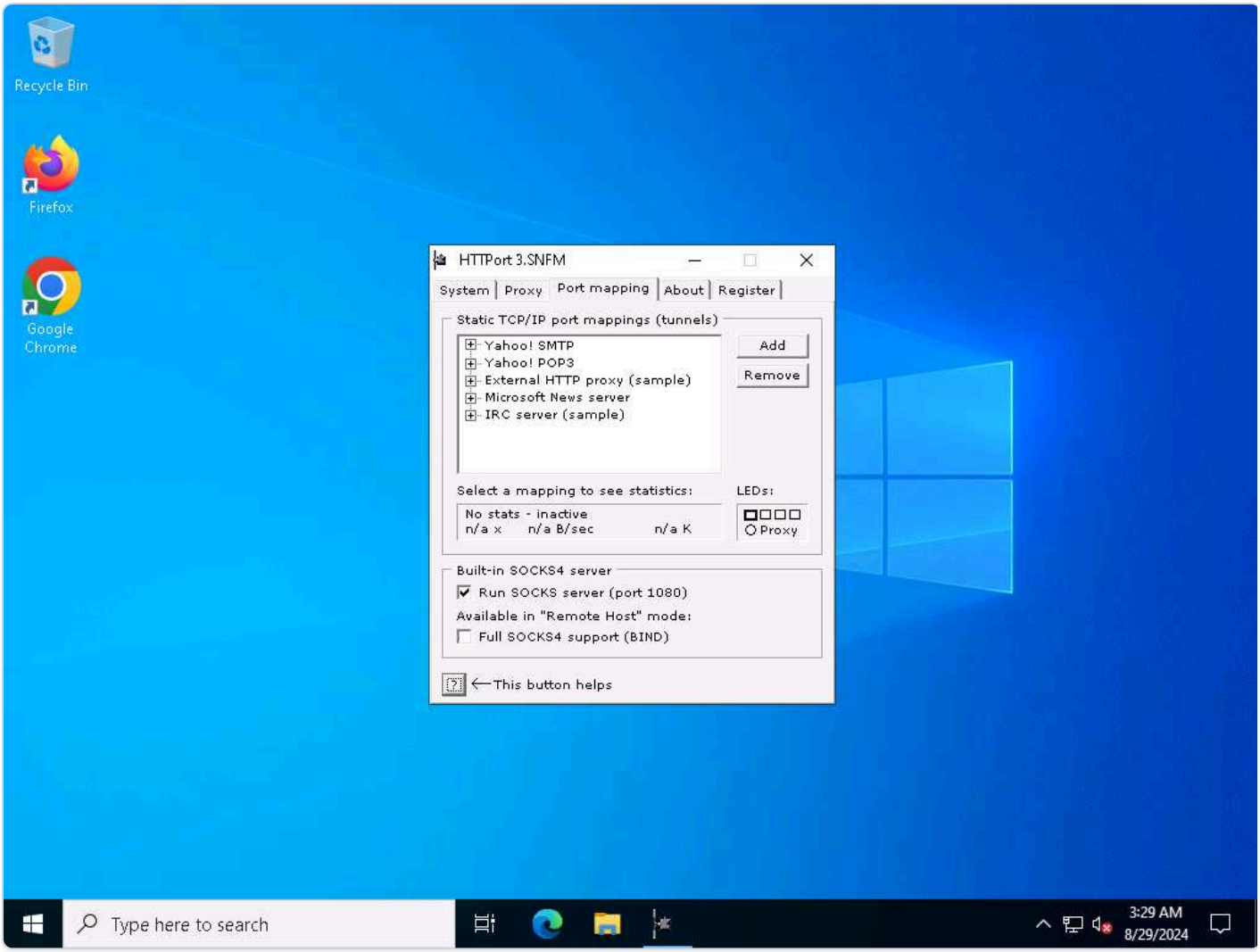
1. Select the **Proxy** tab:
   - Host IP: **172.19.19.9**, Port: **80**
   - Bypass mode: **Remote host**
   - Use personal remote host at: **172.19.19.9**, Port: **80**
   - Password: **magic**

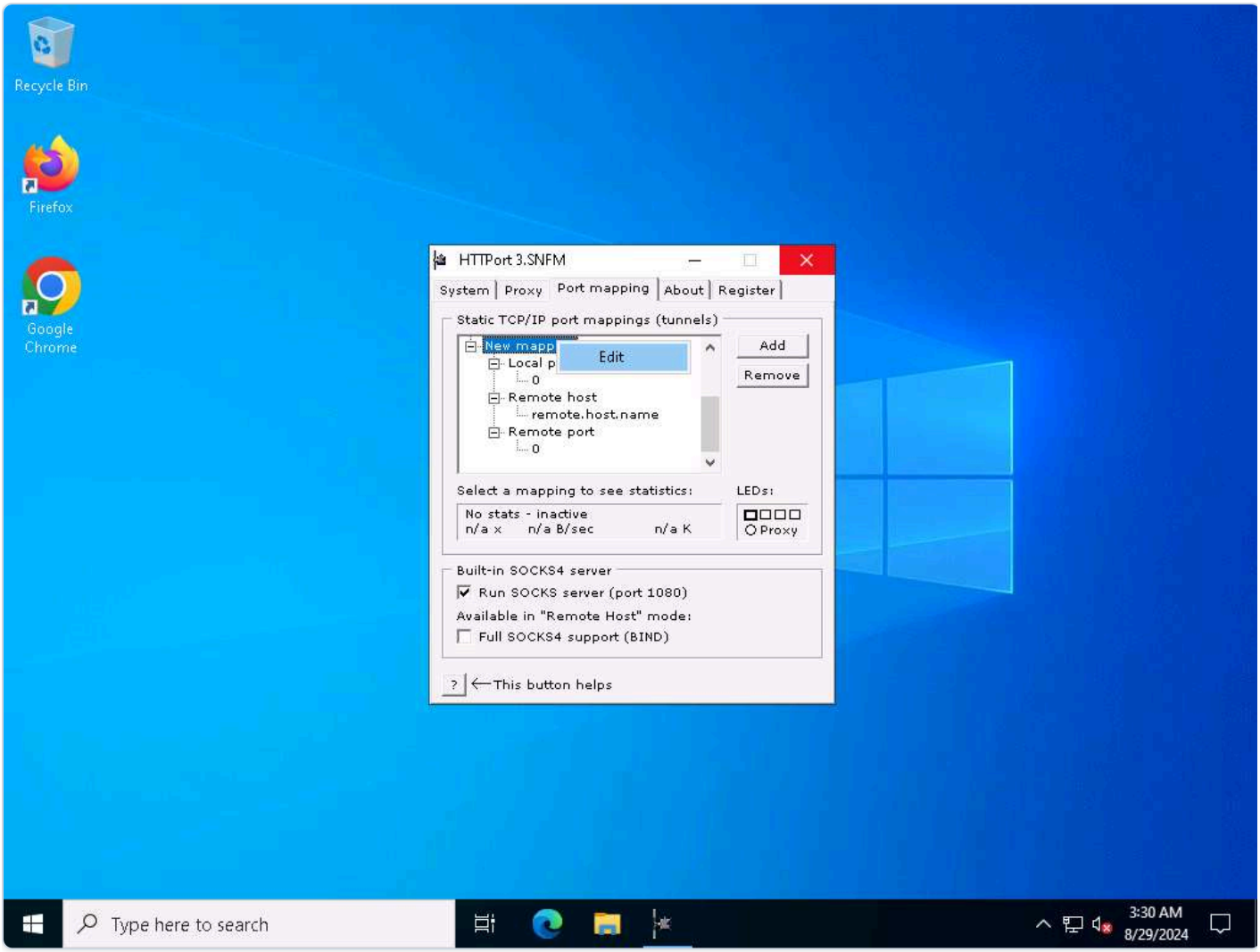


1. Select the **Port mapping** tab $
   ightarrow$ click **Add**.

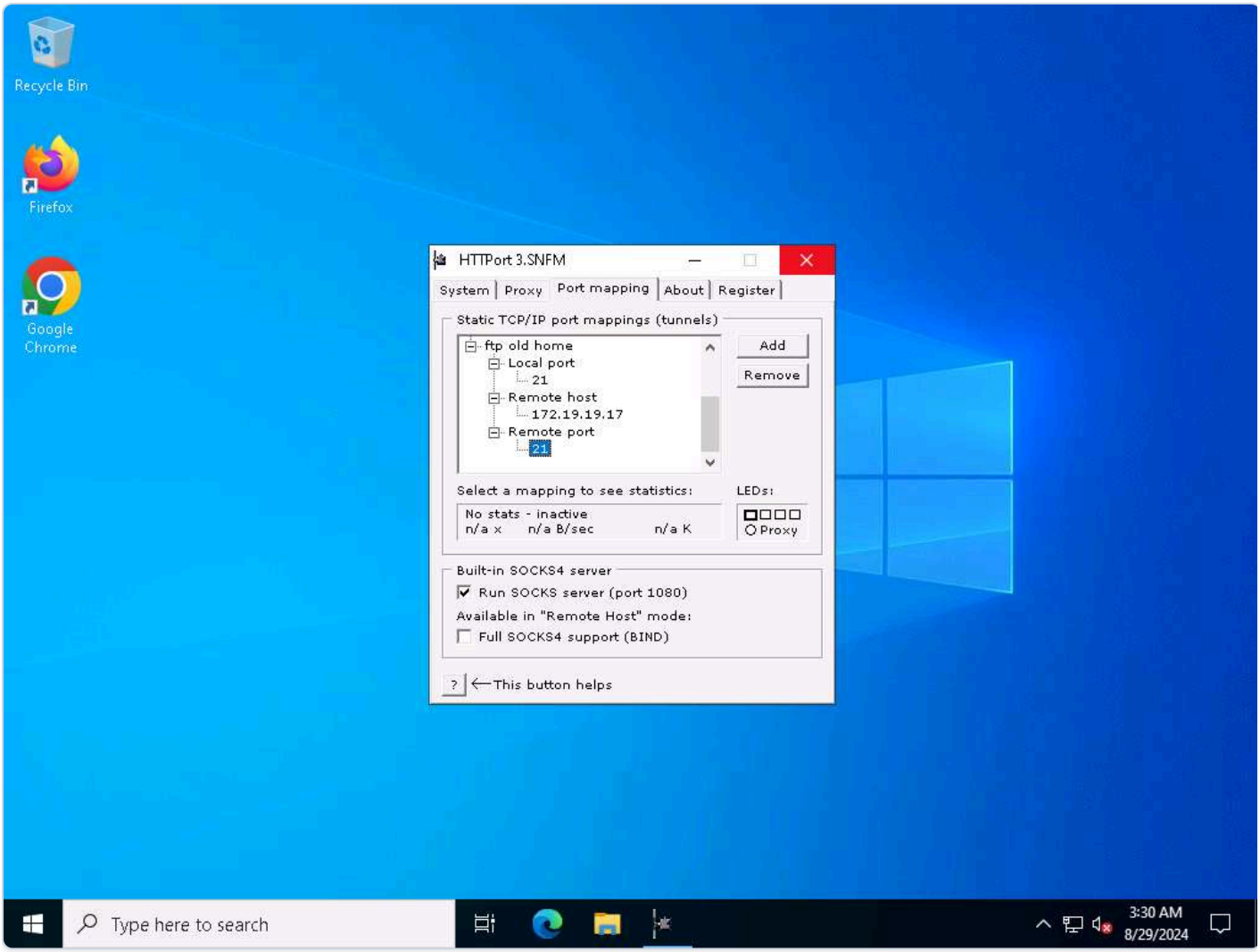


1. Edit mapping node:
   - Name: **ftp old home**
   - Local port: **21**
   - Remote host: **172.19.19.17**
   - Remote port: **21**

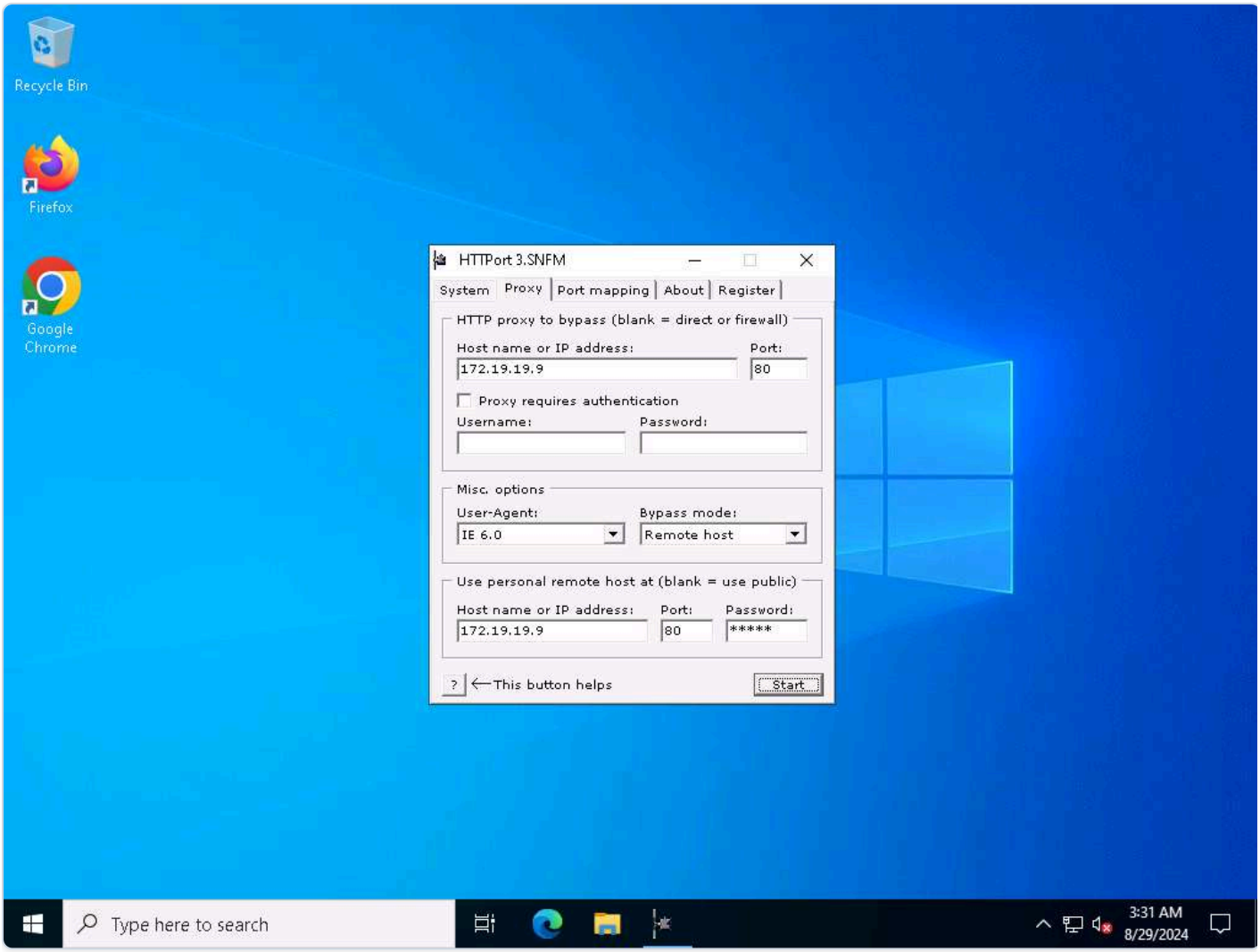

Recycle Bin
Firefox
Google Chrome
HTTPort 3.SNFM
System
Proxy
Port mapping
About
Register
Static TCP/IP port mappings (tunnels)
Yahoo! SMTP
Yahoo! POP3
External HTTP proxy (sample)
Microsoft News server
IRC server (sample)
Add
Remove
Select a mapping to see statistics:
No stats - inactive
n/a x
n/a B/sec
n/a K
LEDs:
Proxy
Built-in SOCKS4 server
Run SOCKS server (port 1080)
Available in "Remote Host" mode:
Full SOCKS4 support (BIND)
This button helps
Type here to search
3:29 AM
8/29/2024



1. Return to the **Proxy** tab and click **Start** to initiate HTTP tunneling.



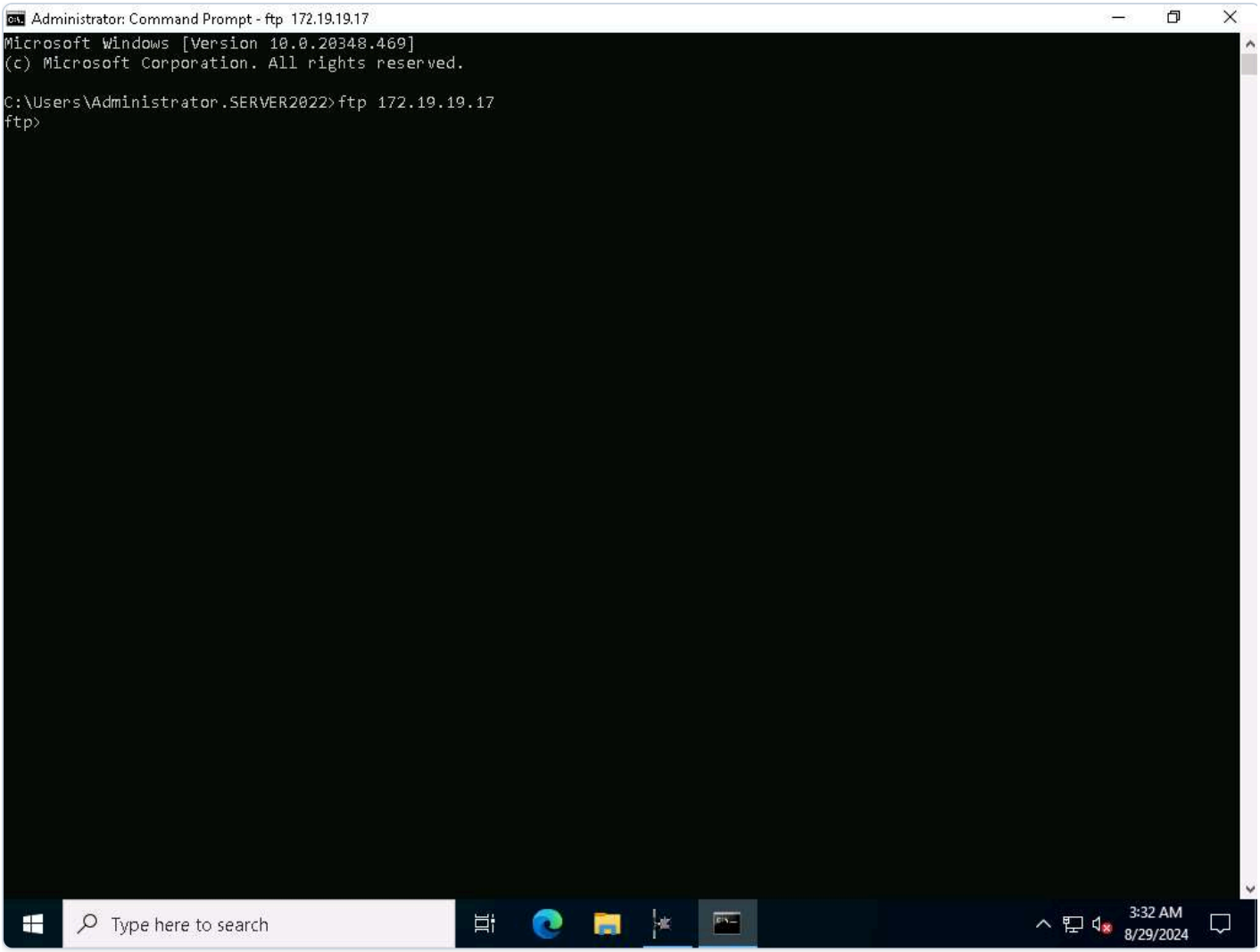
1. Confirm direct connection to 172.19.19.17 remains blocked (**ftp 172.19.19.17**).



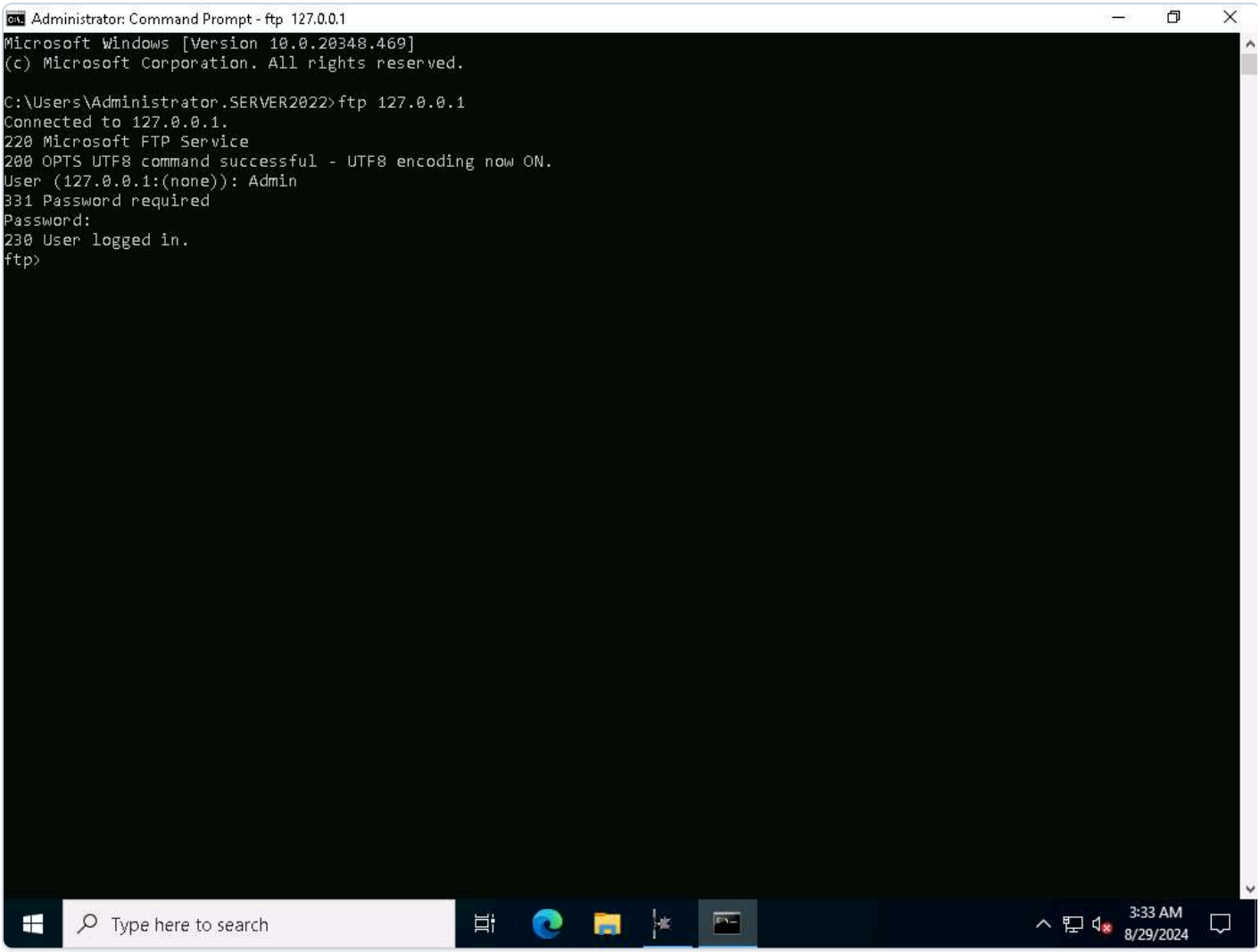
1. Open Command Prompt and connect via local tunnel listener:

   ftp 127.0.0.1

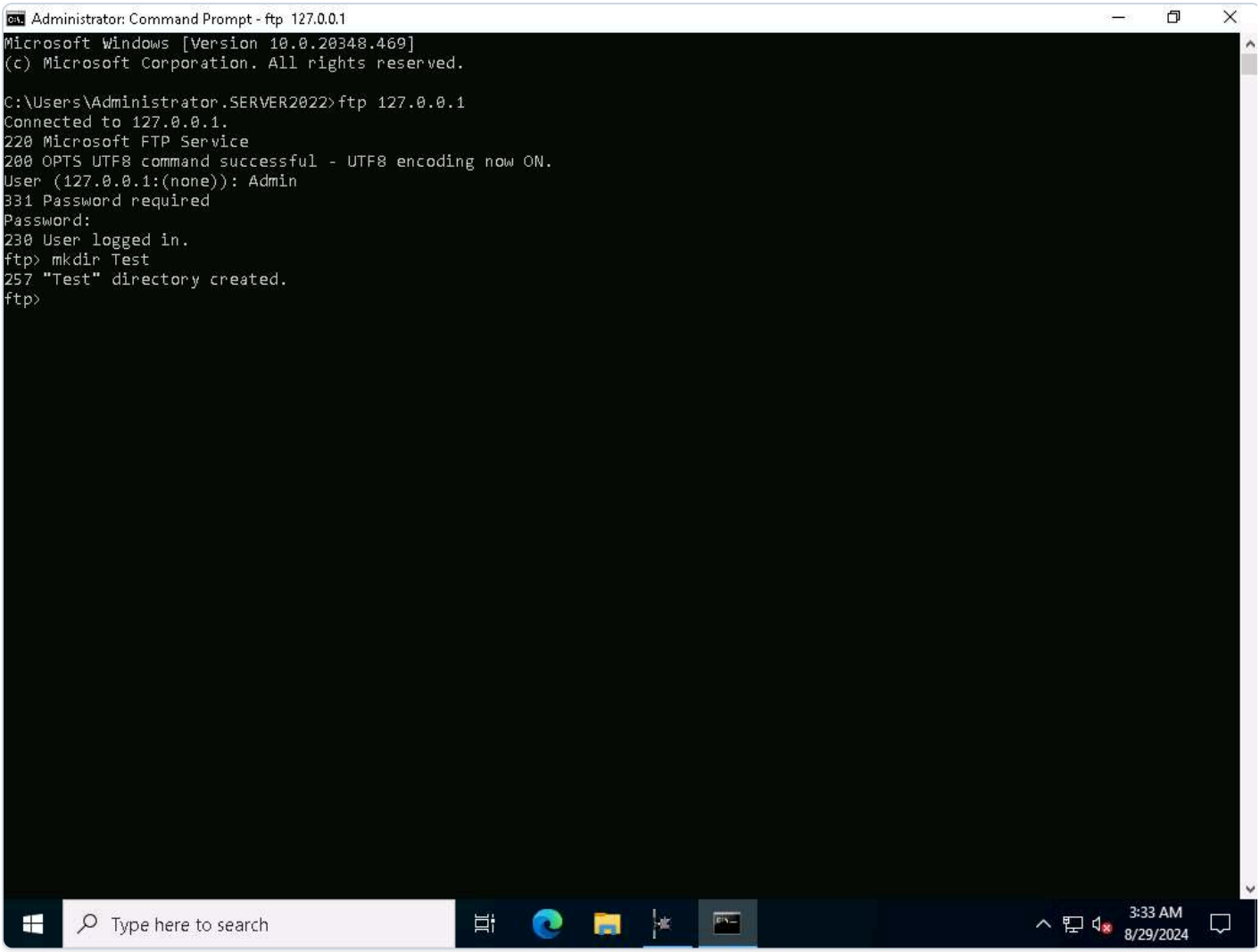
   Log in with username **Admin** and password **test@123**.

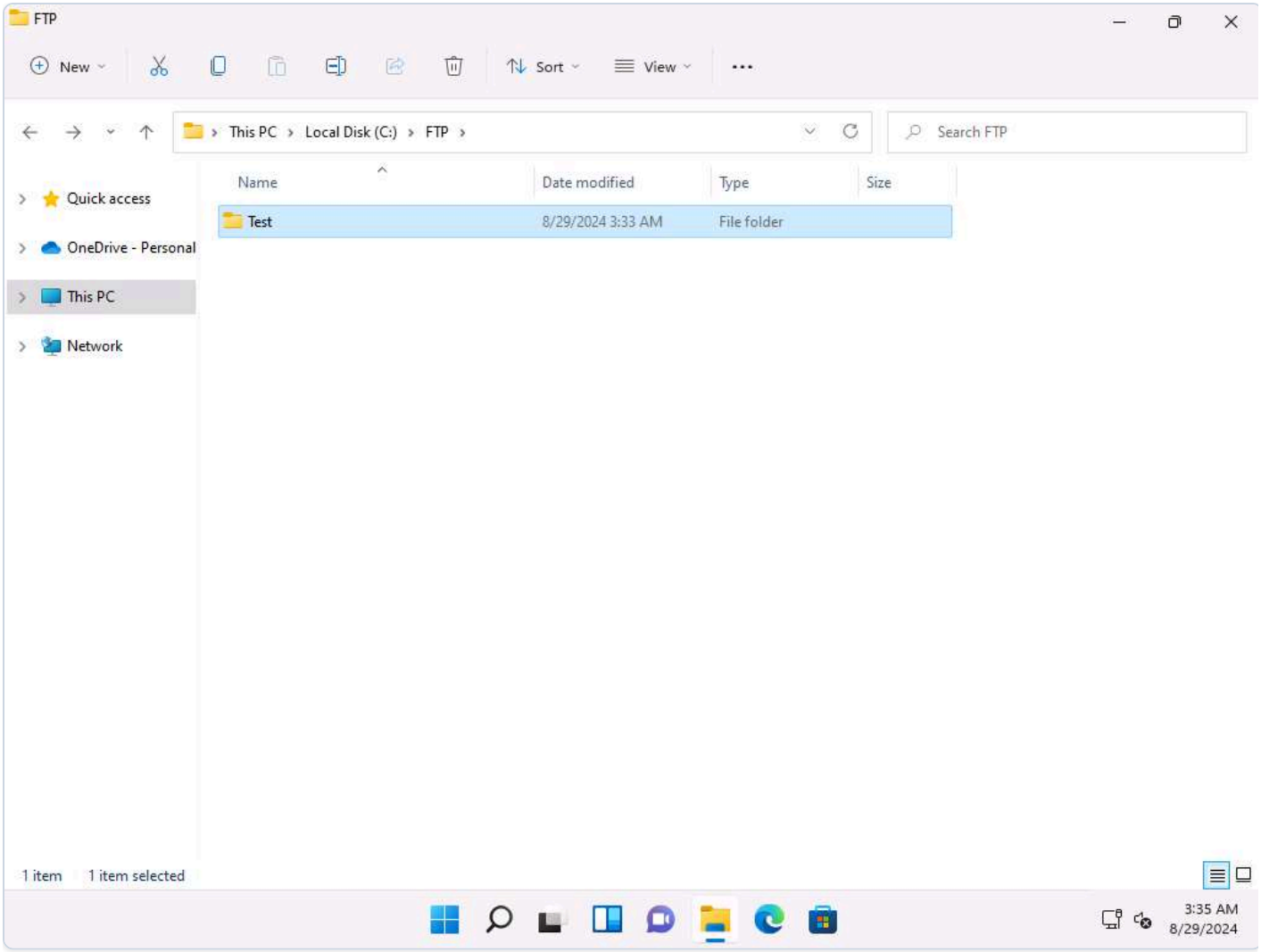


1. Create a directory through the tunneled session:
   mkdir Test



1. Switch to **Sales Department** (**Admin** / **test@123**) and inspect `C:\FTP` to confirm creation of directory `Test`.


Administrator: Command Prompt - ftp 127.0.0.1
Microsoft Windows [Version 10.0.20348.469]
(c) Microsoft Corporation. All rights reserved.
C:\Users\Administrator.SERVER2022>ftp 127.0.0.1
Connected to 127.0.0.1.
220 Microsoft FTP Service
200 OPTS UTF8 command successful - UTF8 encoding now ON.
User (127.0.0.1:(none)): Admin
331 Password required
Password:
230 User logged in.
ftp> mkdir Test
257 "Test" directory created.
ftp>
Type here to search
3:33 AM
8/29/2024



1. Clean up rules and restore service states as instructed. Close all active windows.

This concludes the lab exercise.

IMPORTANT: ❓ **Question 7.3.1.1** On the Windows Server 2022 machine, enable the Firewall and create an outbound rule to block the FTP port. Additionally, configure HTTPort. On the FTP Server machine, utilize the HTTHost server to establish a connection with HTTPort. Then, use HTTP tunneling to access the FTP service on the Sales Department machine. Finally, identify the tab within the HTTHost tool where the Personal Password can be configured. [!SUCCESS] ✅ **Answer:** **options**